

Anchored Pfaffian Hyperplane Sweeps

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1 Theoretical Setup

1.1 Anchored common-chain presentations

Fix integers $N \geq 1$ and $q \geq 0$, a center $c \in \mathbb{R}$, and a half-width $h > 0$. Write

$$\Theta = [c - h, c + h], \quad x = x(\theta) := \frac{\theta - c}{h} \in [-1, 1].$$

The ambient parameter dimension is $p = 1$. For a polynomial S , let $\|\text{coeff}(S)\|_1$ denote the sum of the absolute values of its coefficients in the standard monomial basis.

Let $\eta = (\eta_1, \dots, \eta_q)$ be a common triangular chain on $[-1, 1]$. For $1 \leq j \leq q$, let P_j be its chain polynomial in $(x, y_1, \dots, y_j)$. For $1 \leq i \leq N$, let Q_i be an output polynomial in $(x, y_1, \dots, y_q)$. We use the one-variable convention in (Balcan et al., 2025):

$$M := \max_{1 \leq j \leq q} \deg P_j \quad (q \geq 1), \quad \Delta := \max_{1 \leq i \leq N} \deg Q_i,$$

where degrees are total degrees, and $M = 0$ when $q = 0$. Define

$$G_i(x) := Q_i(x, \eta_1(x), \dots, \eta_q(x)), \quad F_i(\theta) := G_i(x(\theta)),$$

and

$$B_P := \begin{cases} \max_{1 \leq j \leq q} \|\text{coeff}(P_j)\|_1, & q \geq 1, \\ 0, & q = 0, \end{cases} \quad B_Q := \max_{1 \leq i \leq N} \|\text{coeff}(Q_i)\|_1.$$

The affine change from x to θ leaves q, M, Δ unchanged and satisfies $d/d\theta = h^{-1}d/dx$.

For $R > 0$ and $0 < \kappa < \infty$, let $\mathcal{D}_{N,R,\kappa}$ be the class of Borel probability laws on $\mathbb{R}^N$ with a Lebesgue density supported on $[-R, R]^N$ and bounded almost everywhere by κ . No product structure is part of this definition.

1.2 Assumptions

Assumption 1 (Primitive parameter regime). $N \geq 1$, $q \geq 0$, $h > 0$, $R > 0$, and $0 < \kappa < \infty$; $\Theta = [c - h, c + h]$; the class $\mathcal{D}_{N,R,\kappa}$ is nonempty; and all polynomial degrees and coefficient budgets displayed above are finite static presentation data.

Assumption 2 (Balcan common-chain presentation). Each $\eta_j \in C^1([-1, 1])$ satisfies

$$\eta'_j(x) = P_j(x, \eta_1(x), \dots, \eta_j(x)), \quad \deg P_j \leq M,$$

and every output has the common-chain representation

$$G_i(x) = Q_i(x, \eta_1(x), \dots, \eta_q(x)), \quad \deg Q_i \leq \Delta.$$

Thus q is chain length, M is chain degree, Δ is output degree, and $p = 1$.

Assumption 3 (Literal anchor and unit-range certificate). *For every $x \in [-1, 1]$, $|\eta_j(x)| \leq 1$ for all j , and $Q_1 \equiv 1$. No separate lower norm margin or projective-speed bound is assumed.*

Assumption 4 (Arbitrarily correlated capped joint laws). *In every probabilistic clause, α is drawn from an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$. Only cube support and the cap on the full joint density are available; the coordinates may be arbitrarily correlated.*

Assumption 5 (Deterministic affine offset and pivot cover). *Whenever the affine clause is invoked, $F_0 \in C^1(\Theta)$, and the selected measurable sets $E_1, \dots, E_N$ form a partition $I = \bigsqcup_{j=1}^N E_j$ of the interval under consideration, with $F_j(\theta) \neq 0$ on E_j .*

1.3 Objective

Our goal is a single theorem that derives projective conditioning from the anchored presentation, controls central and affine coefficient incidences for arbitrary capped joint laws, recovers the exact monic-polynomial baseline with the leading coefficient kept deterministic, and records the sharp metric scale forced by the two-feature example. The result applies to the declared anchored, unit-range, coefficient-controlled presentation; it makes no claim that every raw Pfaffian presentation admits such a normalization with polynomial budgets.

2 Preliminaries

Write $F = (F_1, \dots, F_N)$ and let I_N be the $N \times N$ identity matrix. Under Assumption 3, define

$$\gamma_F(\theta) := \frac{F(\theta)}{\|F(\theta)\|_2}, \quad \Gamma_{\text{proj}}(F) := \text{ess sup}_{\theta \in \Theta} \|\gamma'_F(\theta)\|_2.$$

The appendix derives both the legality and finiteness of these quantities; no such conclusion is folded into the assumptions.

Set

$$A := (2R)^N \kappa, \quad \mathcal{D} := \mathcal{D}_{N,R,\kappa},$$

and define the ordered all-law, all-interval coefficient

$$C_{\mathcal{D}}^{\text{Pf}}(F; \Theta) := \sup_{\mu \in \mathcal{D}_{N,R,\kappa}} \sup_{\substack{I \subseteq \Theta \text{ interval} \\ |I| > 0}} \frac{\Pr_{\alpha \sim \mu}[\exists \theta \in I : \langle \alpha, F(\theta) \rangle = 0]}{|I|}.$$

Thus the interval supremum is taken for a fixed law before the law supremum.

For the affine interface, given the partition in Assumption 5, index $\beta \in [-R, R]^{N-1}$ by coordinates $i \neq j$ and define

$$T_j(\theta, \beta) := -\frac{F_0(\theta)}{F_j(\theta)} - \sum_{i \neq j} \beta_i \frac{F_i(\theta)}{F_j(\theta)},$$

$$\partial_\theta T_j(\theta, \beta) = -\frac{\left(F'_0(\theta) + \sum_{i \neq j} \beta_i F'_i(\theta)\right) F_j(\theta) - \left(F_0(\theta) + \sum_{i \neq j} \beta_i F_i(\theta)\right) F'_j(\theta)}{F_j(\theta)^2}.$$

All vector and matrix norms below are Euclidean and induced Euclidean norms, respectively; chart velocities use scalar absolute value. We write Leb^m for m -dimensional Lebesgue measure and $\mathcal{H}^m$ for Euclidean Hausdorff measure. The zero-dimensional conventions are $[-R, R]^0 = \{()\}$ and $\text{Leb}^0(\{()\}) = 1$. For an interval I , $|I|$ denotes its Lebesgue length, independently of endpoint convention.

3 Main Theorem

Theorem 3.1 (Anchored coefficient-normalized Pfaffian sweep). *Under Assumptions 1, 2, and 3, the following conclusions hold. The probabilistic conclusions additionally use Assumption 4, and the general affine conclusion also uses Assumption 5.*

Primitive presentation and projective speed. Define

$$D_* := \Delta B_Q(1 + qB_P).$$

For every $x \in [-1, 1]$ and $1 \leq i \leq N$,

$$|G'_i(x)| \leq D_*, \quad \|G'(x)\|_2 \leq \sqrt{N} D_*.$$

Moreover $G_1 = F_1 = 1$, so $\|G(x)\|_2, \|F(\theta)\|_2 \geq 1$. With $\gamma_G = G/\|G\|_2$,

$$\gamma'_G(x) = \frac{(I_N - \gamma_G(x)\gamma_G(x)^\top)G'(x)}{\|G(x)\|_2},$$

$$\gamma_F(\theta) = \gamma_G\left(\frac{\theta - c}{h}\right), \quad \gamma'_F(\theta) = \frac{1}{h}\gamma'_G\left(\frac{\theta - c}{h}\right),$$

and hence

$$\Gamma_{\text{proj}}(F) \leq \frac{\sqrt{N} \Delta B_Q(1 + qB_P)}{h}. \quad (\text{R1})$$

These pointwise statements include one-sided endpoint derivatives. When $q = 0$, $M = B_P = 0$ and $D_ = \Delta B_Q$; dependence on M is degree zero when B_P is fixed. When $N = 1$, or when the Euclidean projector annihilates G' , the normalized curve is stationary and its speed is zero. The case $\Delta = 0$ likewise gives the identity $0 \leq 0$.*

Central swept-hyperplane bound. Fix the deterministic presentation before selecting a law or interval. For every, possibly correlated, $\mu \in \mathcal{D}_{N,R,\kappa}$ and then every positive-length interval $I \subseteq \Theta$,

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : \langle \alpha, F(\theta) \rangle = 0] \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) |I| \leq \frac{AN \Delta B_Q(1 + qB_P)}{\sqrt{2} h} |I|, \quad (\text{R2a})$$

and, with the interval supremum inside the law supremum,

$$C_{\mathcal{D}}^{\text{Pf}}(F; \Theta) \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) \leq \frac{AN \Delta B_Q(1 + qB_P)}{\sqrt{2} h}. \quad (\text{R2b})$$

Before division by $|I|$, the same chain holds for empty and singleton intervals with zero right side and for every endpoint convention. Tangent and multiple roots, stationary normalized pieces, and coefficients for which the combination vanishes identically are included; no simple-root or finite-fiber condition is imposed.

General affine chart bound. For every interval $I \subseteq \Theta$, every measurable pivot partition in Assumption 5, and every, possibly correlated, $\mu \in \mathcal{D}_{N,R,\kappa}$,

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : F_0(\theta) + \langle \alpha, F(\theta) \rangle = 0]$$

$$\leq \kappa \sum_{j=1}^N \int_{E_j} \int_{[-R,R]^{N-1}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta. \quad (\text{R3})$$

This is an inequality in $[0, +\infty]$. It permits a divergent chart integral, arbitrarily small nonzero pivots, cube faces, empty or singleton intervals, included endpoints, tangent and multiple roots, infinite

fibers, $N = 1$, and affine combinations that vanish identically. It assumes neither transversality, a uniform pivot margin, a conditional density, nor coordinate independence.

Exact affine-monic recovery. For every integer $d \geq 1$ and bounded interval $J \subset \mathbb{R}$, choose a nondegenerate $\Theta = [c - h, c + h] \supseteq J$ and set

$$\begin{aligned} F_0(\theta) &= \theta^d, & F_{k+1}(\theta) &= \theta^k \quad (0 \leq k \leq d-1), \\ Q_0(x) &= (c + hx)^d, & Q_{k+1}(x) &= (c + hx)^k \quad (0 \leq k \leq d-1). \end{aligned}$$

Then

$$p_\alpha(\theta) := F_0(\theta) + \langle \alpha, F(\theta) \rangle = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k,$$

with the exact data

$$q = 0, \quad M = 0, \quad B_P = 0, \quad N = d, \quad A = (2R)^d \kappa, \quad \Delta_{\text{aug}} = d.$$

The monic coefficient 1 is deterministic and remains outside the random lower-coefficient vector $\alpha = (\alpha_0, \dots, \alpha_{d-1})$.

For $d \geq 2$, use

$$E_1 = J \cap \{|\theta| \leq 1\}, \quad E_d = J \cap \{|\theta| > 1\}, \quad E_j = \emptyset \quad (j \notin \{1, d\}).$$

The transition points $0, 1, -1$ belong to the constant-pivot cell. The two charts and their literal velocities are

$$T_1(\theta, \beta) = -\theta^d - \sum_{k=1}^{d-1} \beta_k \theta^k, \quad |\partial_\theta T_1| \leq d + R \sum_{k=1}^{d-1} k = d + \frac{Rd(d-1)}{2} \quad (|\theta| \leq 1),$$

$$T_d(\theta, \beta) = -\theta - \sum_{k=0}^{d-2} \beta_k \theta^{k-d+1},$$

$$|\partial_\theta T_d| \leq 1 + R \sum_{k=0}^{d-2} (d-1-k) = 1 + \frac{Rd(d-1)}{2} \leq d + \frac{Rd(d-1)}{2} \quad (|\theta| > 1).$$

For $d = 1$, use $E_1 = J$, the zero-dimensional beta cube, $T_1(\theta) = -\theta$, and $|T'_1(\theta)| = 1$. Consequently, for every possibly correlated $\mu \in \mathcal{D}_{d,R,\kappa}$ and then every bounded J ,

$$\Pr_{\alpha \sim \mu} [\exists \theta \in J : p_\alpha(\theta) = 0] \leq \kappa (2R)^{d-1} \left(d + \frac{Rd(d-1)}{2} \right) |J|. \quad (\text{R4})$$

At $d = 1$ this is exactly $\kappa|J|$. There is no chart-count factor, interval enlargement, law augmentation, conservative remainder, or singular leading coordinate.

Counter-example 1 scale. Let $0 < \delta \leq 1$, $\Theta = [-1, 1]$, $c = 0$, $h = 1$, and

$$G(x) = \left(1, \frac{x}{\delta}\right), \quad F(\theta) = \left(1, \frac{\theta}{\delta}\right).$$

For the uniform law on $[-1, 1]^2$,

$$(h, q, M, \Delta, N, R, \kappa, A, B_P, B_Q) = \left(1, 0, 0, 1, 2, 1, \frac{1}{4}, 1, 0, \frac{1}{\delta}\right),$$

$$\gamma_F(\theta) = \frac{(\delta, \theta)}{\sqrt{\delta^2 + \theta^2}}, \quad \|\gamma'_F(\theta)\|_2 = \frac{\delta}{\delta^2 + \theta^2}, \quad \Gamma_{\text{proj}}(F) = \frac{1}{\delta}.$$

For every $0 < \epsilon \leq \delta$,

$$\Pr[\exists \theta \in [0, \epsilon] : \alpha_1 + \alpha_2 \theta / \delta = 0] = \frac{\epsilon}{4\delta}.$$

Therefore, for $\mathcal{D} = \mathcal{D}_{2,1,1/4}$,

$$\frac{1}{4\delta} \leq C_{\mathcal{D}}^{\text{Pf}}(F; [-1, 1]) \leq \underbrace{A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F)}_{1/\delta} \leq \underbrace{\frac{AN \Delta B_Q (1 + qB_P)}{\sqrt{2} h}}_{\sqrt{2}/\delta}. \quad (\text{R5})$$

This is a lower and upper scale certificate, not an equality or optimality claim for the capacity. It includes $\delta = 1$ and $\epsilon = \delta$ without a limiting argument.

Rate contract. The exposed central variables are $(q, M, \Delta, N, R, \kappa, A, B_P, B_Q, h^{-1})$; the monic variables are $(d, R, \kappa, |J|)$; and the scale-audit variables are (δ, ϵ) . There are no hidden constants or allowed hidden parameter dependencies. The deterministic presentation is fixed before the law and interval ranges. The probability mode is ordinary probability under each fixed full joint law; the horizon mode is the stated per-interval assertion followed, only in (R2b), by the interval supremum and then the law supremum. Central norms and operator norms are Euclidean, affine velocities are scalar, and all coefficient and interval measures use the conventions in Section 2. The admissibility conditions are exactly the numbered assumptions and the local restrictions $d \geq 1$, bounded J , and $0 < \epsilon \leq \delta \leq 1$. No confidence parameter, asymptotic horizon, independence premise, or further normalization claim is present, and every displayed algebraic bridge and numerical factor is literal.

4 Proof Sketch

The argument begins with a coefficient-envelope calculation on the actual unit box. Lemma A.1 bounds polynomial values and partials by their monomial coefficient ℓ_1 budgets, and Proposition A.1 applies the common-chain rule to obtain $D_* = \Delta B_Q (1 + qB_P)$. The literal anchor then supplies a global denominator. Lemma A.2 computes the Euclidean projector derivative, and Proposition A.2 transfers it through $x = (\theta - c)/h$ with the exact factor h^{-1} .

For the central event, Lemma A.3 computes the two tangential Jacobians of the incidence hypersurface. The area and coarea formulas (Federer, 1969) turn projection multiplicity into the coefficient-volume inequality in Proposition A.3; the null classes and boundary regimes are handled in Lemma A.4. Ball's sharp central cube-section theorem (Ball, 1986) is scaled exactly in Lemma A.5. Proposition A.4 then uses the single full joint-density cap, and Proposition A.5 performs the visible speed substitution and the interval-then-law supremum closure.

The affine argument solves the root equation in a selected nonzero coefficient coordinate. Lemmas A.7 and A.8 establish finite-level Lipschitz legality and the exact determinant $|\partial_\theta T_j|$. Proposition A.6 applies the equal-dimensional area formula, while Lemma A.10 proves that every root enters a least finite pivot level. Monotone convergence yields the extended-real bound in Proposition A.7, without transversality or a uniform pivot margin.

For monic polynomials, Proposition A.8 constructs the exact normalized presentation while keeping the leading coefficient in the deterministic offset. Lemma A.11 supplies the prescribed two cells, and Lemmas A.12 and A.13 compute their velocities. The general affine theorem is transferred without loss in Proposition A.9; Lemma A.14 keeps the sharper high-chart constant until the final coefficient-one comparison, while Proposition A.10 handles the zero-dimensional $d = 1$ case.

Finally, Proposition A.12 computes the exact projective speed of the two-feature example, Lemma A.15 identifies both closed coefficient wedges and their area, and Proposition A.13 compares the resulting $1/(4\delta)$ certificate with the distinct projective and raw-presentation upper scales. Proposition A.14 places these exact interfaces under their respective numbered assumptions, after which the main theorem is their direct conjunction.

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A Proof Details

A.1 Unit-box coefficient envelopes and chain derivatives

Lemma A.1 (Literal unit-box coefficient envelope). *Let $m \geq 1$ and $d \geq 0$ be integers, and let*

$$S(z) = \sum_{\nu \in \mathcal{A}} a_\nu z^\nu, \quad \mathcal{A} \subseteq \{\nu \in \mathbb{N}_0^m : |\nu|_1 \leq d\}$$

have finite coefficient support. Then, for every $z \in [-1, 1]^m$,

$$|S(z)| \leq \|\text{coeff}(S)\|_1, \quad |\partial_{z_r} S(z)| \leq d \|\text{coeff}(S)\|_1 \quad (1 \leq r \leq m).$$

These statements include $d = 0$, constant polynomials, and the zero polynomial without assigning a degree to the zero polynomial.

Proof. For $z \in [-1, 1]^m$, every monomial satisfies $|z^\nu| \leq 1$, and hence

$$|S(z)| \leq \sum_{\nu \in \mathcal{A}} |a_\nu| |z^\nu| \leq \sum_{\nu \in \mathcal{A}} |a_\nu| = \|\text{coeff}(S)\|_1.$$

For a fixed coordinate r , termwise differentiation gives

$$\partial_{z_r} S(z) = \sum_{\substack{\nu \in \mathcal{A} \\ \nu_r \geq 1}} \nu_r a_\nu z^{\nu - e_r}.$$

Since $\nu_r \leq |\nu|_1 \leq d$ and $|z^{\nu - e_r}| \leq 1$,

$$\begin{aligned} |\partial_{z_r} S(z)| &\leq \sum_{\substack{\nu \in \mathcal{A} \\ \nu_r \geq 1}} \nu_r |a_\nu| |z^{\nu - e_r}| \\ &\leq d \sum_{\substack{\nu \in \mathcal{A} \\ \nu_r \geq 1}} |a_\nu| \leq d \|\text{coeff}(S)\|_1. \end{aligned}$$

No monomial-count factor occurs. If $d = 0$, the differentiated sum is empty; if $S = 0$, both inequalities are $0 \leq 0$. $\square$

Proposition A.1 (Coordinate derivative envelope). *Under Assumptions 2 and 3, define*

$$D_* := \Delta B_Q (1 + q B_P).$$

Then $G_i \in C^1([-1, 1])$ and

$$|G'_i(x)| \leq D_* \quad (x \in [-1, 1], 1 \leq i \leq N).$$

For $q = 0$, this is $|G'_i| \leq \Delta B_Q$ with $M = B_P = 0$. For fixed B_P , the bound has degree-zero dependence on M .

Proof. Fix $x \in [-1, 1]$ and $i \leq N$, and write

$$z(x) := (x, \eta_1(x), \dots, \eta_q(x)).$$

When P_j is evaluated at $z(x)$ it ignores later chain coordinates, as in Assumption 2. The unit-range condition gives $z(x) \in [-1, 1]^{q+1}$; for $q = 0$, this says $z(x) = (x) \in [-1, 1]$.

Because $\deg Q_i \leq \Delta$, Lemma A.1 gives

$$|\partial_x Q_i(z(x))| \leq \Delta B_Q, \quad |\partial_{y_j} Q_i(z(x))| \leq \Delta B_Q \quad (1 \leq j \leq q).$$

For $q \geq 1$, the value part of that lemma and the definition of B_P give

$$|P_j(z(x))| \leq \|\text{coeff}(P_j)\|_1 \leq B_P.$$

The constant is one because every monomial has magnitude at most one on the actual unit box; neither M nor a monomial count is introduced.

The multivariable chain rule (Rudin, 1976) and the chain equations yield

$$\begin{aligned} G'_i(x) &= \partial_x Q_i(z(x)) + \sum_{j=1}^q \partial_{y_j} Q_i(z(x)) \eta'_j(x) \\ &= \partial_x Q_i(z(x)) + \sum_{j=1}^q \partial_{y_j} Q_i(z(x)) P_j(z(x)). \end{aligned}$$

Consequently,

$$\begin{aligned} |G'_i(x)| &\leq |\partial_x Q_i(z(x))| + \sum_{j=1}^q |\partial_{y_j} Q_i(z(x))| |P_j(z(x))| \\ &\leq \Delta B_Q + \sum_{j=1}^q \Delta B_Q B_P \\ &= \Delta B_Q + q \Delta B_Q B_P = \Delta B_Q (1 + q B_P) = D_*. \end{aligned}$$

The calculation holds on the interval interior. Every term is continuous on $[-1, 1]$, so the same identity and estimate hold at the endpoints by the corresponding one-sided limits.

When $q = 0$, both sums are empty and

$$G'_i(x) = Q'_i(x), \quad |G'_i(x)| \leq \Delta B_Q.$$

If $\Delta = 0$, every output is constant and both sides vanish. A constant or zero Q_i has zero derivative even when the global Δ is positive. If $M = 0$ with $q \geq 1$, each nonzero P_j is constant and the same value bound applies; a zero P_j , or $B_P = 0$, removes its chain contribution. These cases introduce no additional convention or constant. $\square$

A.2 Euclidean normalization and projective speed

Lemma A.2 (Derivative of a normalized Euclidean curve). *Let $J = [a, b]$ be nondegenerate and let $v \in C^1(J; \mathbb{R}^N)$ satisfy $v(t) \neq 0$ on J . Define*

$$r(t) := \|v(t)\|_2, \quad \gamma(t) := \frac{v(t)}{r(t)}.$$

Then $r, \gamma \in C^1(J)$ under the one-sided endpoint convention, and

$$r'(t) = \gamma(t)^\top v'(t), \quad \gamma'(t) = \frac{(I_N - \gamma(t)\gamma(t)^\top)v'(t)}{\|v(t)\|_2}.$$

The matrix $I_N - \gamma\gamma^\top$ is the Euclidean orthogonal projector onto $\gamma^\perp$, so

$$\|I_N - \gamma(t)\gamma(t)^\top\|_{2 \rightarrow 2} \leq 1, \quad \|\gamma'(t)\|_2 \leq \frac{\|v'(t)\|_2}{\|v(t)\|_2}.$$

For $N = 1$, the projector is zero and γ is constant.

Proof. Nonvanishing gives $r > 0$. Differentiating $r(t)^2 = v(t)^\top v(t)$ gives

$$2r(t)r'(t) = 2v(t)^\top v'(t), \quad r'(t) = \frac{v(t)^\top v'(t)}{r(t)} = \gamma(t)^\top v'(t).$$

Differentiating v/r therefore yields

$$\begin{aligned} \gamma'(t) &= \frac{v'(t)}{r(t)} - \frac{v(t)r'(t)}{r(t)^2} \\ &= \frac{v'(t) - \gamma(t)\gamma(t)^\top v'(t)}{r(t)} \\ &= \frac{(I_N - \gamma(t)\gamma(t)^\top)v'(t)}{\|v(t)\|_2}. \end{aligned}$$

The right sides are continuous, so the interior identities extend to the endpoints by one-sided limits.

Put $P = I_N - \gamma\gamma^\top$. Since $\|\gamma\|_2 = 1$,

$$P^\top = P, \quad P^2 = I_N - 2\gamma\gamma^\top + \gamma(\gamma^\top\gamma)\gamma^\top = P.$$

Also $P\gamma = 0$, while $Pz = z$ for $z \perp \gamma$, so P is the stated orthogonal projector. For every $z \in \mathbb{R}^N$,

$$\|Pz\|_2^2 = z^\top P^2 z = \|z\|_2^2 - (\gamma^\top z)^2 \leq \|z\|_2^2.$$

This proves the contraction and the derivative bound. If $N = 1$, continuity and nonvanishing prevent the unit scalar γ from changing sign, and algebraically $P = 1 - \gamma^2 = 0$. $\square$

Proposition A.2 (Anchored projective-speed certificate). *Under Assumptions 1 and 3, and Proposition A.1, define*

$$D_* := \Delta B_Q(1 + qB_P), \quad \gamma_G(x) := \frac{G(x)}{\|G(x)\|_2}.$$

Then for every $x \in [-1, 1]$ and $\theta \in \Theta$,

$$G_1(x) = F_1(\theta) = 1, \quad \|G(x)\|_2, \|F(\theta)\|_2 \geq 1.$$

Both normalized curves are globally C^1 , and

$$\begin{aligned} \gamma'_G(x) &= \frac{(I_N - \gamma_G(x)\gamma_G(x)^\top)G'(x)}{\|G'(x)\|_2}, \\ \gamma_F(\theta) &= \gamma_G\left(\frac{\theta - c}{h}\right), \quad \gamma'_F(\theta) = \frac{1}{h}\gamma'_G\left(\frac{\theta - c}{h}\right). \end{aligned}$$

In particular,

$$\Gamma_{\text{proj}}(F) \leq \frac{\sqrt{N} D_*}{h} = \frac{\sqrt{N} \Delta B_Q(1 + qB_P)}{h}.$$

Proof. The literal anchor gives

$$G_1(x) = Q_1(x, \eta_1(x), \dots, \eta_q(x)) = 1, \quad F_1(\theta) = G_1(x(\theta)) = 1.$$

Thus

$$\|G(x)\|_2^2 = 1 + \sum_{i=2}^N |G_i(x)|^2 \geq 1, \quad \|F(\theta)\|_2^2 = 1 + \sum_{i=2}^N |F_i(\theta)|^2 \geq 1,$$

with empty sums when $N = 1$. Nonvanishing is therefore proved before Lemma A.2 is applied to $v = G$.

Proposition A.1 gives $|G'_i(x)| \leq D_*$ for all i . Since $D_* \geq 0$,

$$\|G'(x)\|_2^2 = \sum_{i=1}^N |G'_i(x)|^2 \leq \sum_{i=1}^N D_*^2 = ND_*^2, \quad \|G'(x)\|_2 \leq \sqrt{N} D_*.$$

Lemma A.2 supplies the displayed projector identity, whose denominator is at least one and whose numerator projector has operator norm at most one. Hence

$$\|\gamma'_G(x)\|_2 \leq \frac{\|G'(x)\|_2}{\|G(x)\|_2} \leq \sqrt{N} D_*.$$

Since $F(\theta) = G(x(\theta))$, normalization commutes exactly with evaluation:

$$\gamma_F(\theta) = \frac{G(x(\theta))}{\|G(x(\theta))\|_2} = \gamma_G(x(\theta)).$$

The vector chain rule (Rudin, 1976) and $x'(\theta) = h^{-1}$ give

$$\gamma'_F(\theta) = \gamma'_G(x(\theta))x'(\theta) = \frac{1}{h}\gamma'_G\left(\frac{\theta - c}{h}\right),$$

including one-sided endpoints by continuity. Therefore

$$\|\gamma'_F(\theta)\|_2 \leq \frac{\sqrt{N} D_*}{h},$$

and taking the defining essential supremum proves the result.

If $N = 1$, the anchor forces $F = G = (1)$, so the projector and speed are zero; also $\Delta = D_* = 0$. If $q = 0$, then $M = B_P = 0$ and the rate becomes $\sqrt{N}\Delta B_Q/h$. If the projector annihilates G' , both normalized derivatives vanish. For $G(x) = (1, x/\delta)$ with $0 < \delta \leq 1$, one has

$$B_Q = \frac{1}{\delta}, \quad D_* = \frac{1}{\delta}, \quad \gamma_G(x) = \frac{(1, x/\delta)}{\sqrt{1 + (x/\delta)^2}}, \quad \|\gamma'_G(x)\|_2 = \frac{1/\delta}{1 + (x/\delta)^2}.$$

Thus its exact speed supremum is $1/\delta$, while the general certificate is $\sqrt{2}/\delta$; both preserve the required scale. $\square$

A.3 Central incidence geometry and coefficient volume

Lemma A.3 (Tangential Jacobians of the central incidence). *Under Assumption 1 and Proposition A.2, let $J \subseteq \text{int}(\Theta)$ be open, put $K^\circ = (-R, R)^N$, and write $\gamma = \gamma_F$. Define*

$$\mathcal{S}_J := \{(\theta, a) \in J \times K^\circ : \langle a, \gamma(\theta) \rangle = 0\}, \quad \pi(\theta, a) := a, \quad \tau(\theta, a) := \theta.$$

Then $\mathcal{S}_J$ is an embedded C^1 , countably N -rectifiable hypersurface. With

$$u(\theta, a) := \langle a, \gamma'(\theta) \rangle, \quad n(\theta, a) := \frac{(u(\theta, a), \gamma(\theta))}{\sqrt{1 + u(\theta, a)^2}},$$

n is a unit normal and

$$J_{\mathcal{S}_J} \pi = \frac{|u|}{\sqrt{1 + u^2}}, \quad J_{\mathcal{S}_J} \tau = \frac{1}{\sqrt{1 + u^2}}.$$

Consequently, including at $u = 0$,

$$J_{\mathcal{S}_J} \pi = |\langle a, \gamma'_F(\theta) \rangle| J_{\mathcal{S}_J} \tau. \tag{A.1}$$

Proof. On $U = J \times \mathbb{R}^N$, define

$$g(\theta, a) := \langle a, \gamma(\theta) \rangle.$$

Proposition A.2 gives $\gamma \in C^1$ and $\|\gamma\|_2 = 1$, so

$$\nabla g(\theta, a) = (\langle a, \gamma'(\theta) \rangle, \gamma(\theta)) = (u, \gamma), \quad \|\nabla g(\theta, a)\|_2^2 = 1 + u^2 \geq 1.$$

At each incidence, some coefficient component $\gamma_k(\theta)$ is nonzero, and $\partial g / \partial a_k = \gamma_k(\theta)$. The Euclidean C^1 implicit-function theorem (Rudin, 1976) therefore writes the zero set locally as a C^1 graph over the other N coordinates. Thus it is an embedded C^1 hypersurface with tangent space $\ker Dg$ and unit normal $\nabla g / \|\nabla g\|_2$, exactly at the regularity proved above. No root-transversality premise is needed: tangent roots $u = 0$ remain regular because the coefficient gradient is the unit vector γ .

Let $e_0 = (1, 0, \dots, 0) \in \mathbb{R}^{N+1}$ be the θ -direction and $T = T_{(\theta, a)}\mathcal{S}_J = n^\perp$. Choose an orthonormal basis $v_1, \dots, v_N$ of T and set $s_i = \langle v_i, e_0 \rangle$. Coefficient projection deletes only the e_0 component, so

$$\langle \pi v_i, \pi v_j \rangle = \delta_{ij} - s_i s_j.$$

The determinant of this rank-one perturbation and the orthonormal basis $v_1, \dots, v_N, n$ give

$$(J_{\mathcal{S}_J} \pi)^2 = \det(I_N - ss^\top) = 1 - \|s\|_2^2,$$

$$1 = \|e_0\|_2^2 = \sum_{i=1}^N \langle e_0, v_i \rangle^2 + \langle e_0, n \rangle^2 = \|s\|_2^2 + \langle e_0, n \rangle^2.$$

Therefore

$$J_{\mathcal{S}_J} \pi = |\langle e_0, n \rangle| = \frac{|u|}{\sqrt{1 + u^2}}.$$

The tangential gradient of the scalar coordinate τ is $P_T e_0$, and hence

$$J_{\mathcal{S}_J} \tau = \|P_T e_0\|_2 = \sqrt{1 - \langle e_0, n \rangle^2} = \frac{1}{\sqrt{1 + u^2}}.$$

This quantity is strictly positive for every finite a . Multiplication by $|u|$ proves (A.1) without division by u or selection of a root branch. $\square$

Lemma A.4 (Null boundary and degenerate central classes). *Under Assumptions 1 and 3, and Proposition A.2, let $I \subseteq \Theta$ be an interval and write $\gamma = \gamma_F$. Then:*

1. For every a and $\theta \in I$,

$$\langle a, F(\theta) \rangle = 0 \iff \langle a, \gamma(\theta) \rangle = 0. \tag{A.2}$$

2. Coefficients producing a root in $I \setminus \text{int}(I)$ lie in at most two proper linear hyperplanes and are Lebesgue-null.

3. If $I \neq \emptyset$, the set

$$\mathcal{Z}_I := \{a : \langle a, F(\theta) \rangle = 0 \text{ for every } \theta \in I\}$$

is a proper null linear subspace and has probability zero under every absolutely continuous law.

4. If $L \subseteq \Theta$ is nondegenerate and $\gamma' = 0$ on its interior, then γ is constant on L ; the root coefficients over L form one proper hyperplane while the sweep integrand vanishes.
5. If $N = 1$, then $F = \gamma = (1)$, the root coefficient set is $\{0\}$, and both sides of Proposition A.3 below are zero for every interval.

Proof. Proposition A.2 gives $F = \|F\|_2 \gamma$ with $\|F\|_2 > 0$, proving (A.2).

An interval differs from its ordinary interior by at most two endpoints. At a fixed endpoint t , the root coefficients form

$$H_t := \{a : \langle a, \gamma(t) \rangle = 0\} = \gamma(t)^\perp.$$

This hyperplane is proper because $\|\gamma(t)\|_2 = 1$. An orthogonal change of coordinates sends it to $\{a_1 = 0\}$, which is null by Fubini's theorem (Bogachev, 2007); orthogonal maps preserve Lebesgue measure, and a finite union remains null.

If $I \neq \emptyset$, choose $t_0 \in I$. The set $\mathcal{Z}_I$ is an intersection of kernels and hence a linear subspace, with

$$\mathcal{Z}_I \subseteq \{a : \langle a, F(t_0) \rangle = 0\}.$$

The containing hyperplane is proper because $F_1(t_0) = 1$. Thus $\mathcal{Z}_I$ is null, and absolute continuity gives its probability-zero consequence, including the zero coefficient vector.

If $\gamma' = 0$ on the interior of L , the scalar mean-value theorem (Rudin, 1976) applied coordinate-wise makes γ constant there, and continuity extends that value to included endpoints. The root set is then one fixed hyperplane and $\langle a, \gamma' \rangle = 0$ throughout the interval interior.

For $N = 1$, the anchor gives $F = \gamma = (1)$. A root on a nonempty interval exists only for $a = 0$; on an empty interval none exists. The coefficient volume is zero. Also $\gamma' = 0$ and $\gamma(\theta)^\perp \cap [-R, R] = \{0\}$; although $\mathcal{H}^0(\{0\}) = 1$, the section integrand is zero. $\square$

Proposition A.3 (Central root-set coefficient-volume inequality). *Under Assumptions 1 and 3, Proposition A.2, and Lemmas A.3 and A.4, every interval $I \subseteq \Theta$, with any endpoint convention, satisfies*

$$\begin{aligned} & \text{Leb}^N \{a \in [-R, R]^N : \exists \theta \in I, \langle a, F(\theta) \rangle = 0\} \\ & \leq \int_I \int_{\gamma_F(\theta)^\perp \cap [-R, R]^N} |\langle \gamma'_F(\theta), a \rangle| d\mathcal{H}^{N-1}(a) d\theta. \end{aligned} \tag{A.3}$$

The bound requires neither simple roots nor transversality. Tangent incidences have null coefficient image, distinct roots increase projection multiplicity, and identically zero combinations may have infinite multiplicity only on the null subspace in Lemma A.4.

Proof. Lemma A.4 proves the assertion for $N = 1$, so assume $N \geq 2$. Put

$$K = [-R, R]^N, \quad K^\circ = (-R, R)^N, \quad J = \text{int}(I), \quad \gamma = \gamma_F.$$

If $J = \emptyset$, then I is empty or a singleton; its root set is empty or a proper hyperplane, and both the coefficient volume and outer integral are zero. Suppose $J \neq \emptyset$.

The root coefficient set is Borel. Indeed, choose compact intervals $J_m \subset J$ with $J_m \uparrow J$. The zero set of the continuous map $(\theta, a) \mapsto \langle a, \gamma(\theta) \rangle$ in $J_m \times K$ is compact, so its coefficient projection is compact. The interior-root set is the countable union of these projections; adjoining at most two endpoint hyperplanes preserves Borel measurability.

The cube boundary is Lebesgue-null, and endpoint root classes are null by Lemma A.4. The left side of (A.3) therefore has the same volume as $\pi(\mathcal{S}_J)$. For $a \in \mathbb{R}^N$, define the extended-valued multiplicity

$$\mathcal{N}_J(a) := \#\{\theta \in J : (\theta, a) \in \mathcal{S}_J\}.$$

The indicator of a nonempty fiber is at most $\mathcal{N}_J(a)$, even for an infinite fiber. The area formula for rectifiable sets (Federer, 1969) applies because Lemma A.3 supplies a countably rectifiable

hypersurface and π is 1-Lipschitz. Hence, in the extended nonnegative sense,

$$\begin{aligned}\text{Leb}^N(\pi(\mathcal{S}_J)) &= \int_{\mathbb{R}^N} \mathbf{1}_{\{\mathcal{N}_J(a) \geq 1\}} da \\ &\leq \int_{\mathbb{R}^N} \mathcal{N}_J(a) da \\ &= \int_{\mathcal{S}_J} J_{\mathcal{S}_J} \pi(\theta, a) d\mathcal{H}^N(\theta, a).\end{aligned}\tag{A.4}$$

Using (A.1) and then the coarea formula on the rectifiable hypersurface (Federer, 1969), with $\tau(\theta, a) = \theta$, transforms the final integral into

$$\begin{aligned}&\int_{\mathcal{S}_J} |\langle a, \gamma'(\theta) \rangle| J_{\mathcal{S}_J} \tau(\theta, a) d\mathcal{H}^N(\theta, a) \\ &= \int_J \int_{\gamma(\theta)^\perp \cap K^\circ} |\langle \gamma'(\theta), a \rangle| d\mathcal{H}^{N-1}(a) d\theta.\end{aligned}\tag{A.5}$$

The fiber identification is exact: $a \mapsto (\theta, a)$ is an isometry from $\gamma(\theta)^\perp \cap K^\circ$ onto the corresponding τ -fiber. The coarea hypotheses are met because (A.1) gives the measurable nonnegative integrand and $J_{\mathcal{S}_J} \tau > 0$.

Replacing the open cube section in (A.5) by the closed one changes no $\mathcal{H}^{N-1}$ integral. The cube boundary is the union of $2N$ affine faces. A central hyperplane $\gamma(\theta)^\perp$ cannot contain a face: containment would force its tangential normal components to vanish and its remaining component times $\pm R$ to be zero, contradicting $R > 0$ and $\|\gamma(\theta)\|_2 = 1$. Its intersection with each face has dimension at most $N - 2$ and is therefore $\mathcal{H}^{N-1}$ -null. Replacing J by I also changes no outer integral because at most two points are added. Equations (A.4)–(A.5) prove (A.3).

To verify the root-type scope, define

$$\mathcal{C}_J := \{(\theta, a) \in \mathcal{S}_J : \langle a, \gamma'(\theta) \rangle = 0\}.$$

On this measurable set, Lemma A.3 gives $J_{\mathcal{S}_J} \pi = 0$. A second area-formula application gives

$$\int_{\mathbb{R}^N} \#(\mathcal{C}_J \cap \pi^{-1}(a)) da = \int_{\mathcal{C}_J} J_{\mathcal{S}_J} \pi d\mathcal{H}^N = 0,$$

so $\text{Leb}^N(\pi(\mathcal{C}_J)) = 0$. At a root,

$$\frac{d}{d\theta} \langle a, F(\theta) \rangle = \|F(\theta)\|_2 \langle a, \gamma'(\theta) \rangle,$$

because the term containing $\langle a, \gamma(\theta) \rangle$ vanishes. Thus tangent and multiple roots lie in this null image. Distinct roots add preimages and only increase $\mathcal{N}_J$. An identically zero combination has infinite multiplicity but lies in the proper null subspace from Lemma A.4. Endpoints and stationary pieces were also handled there. No finite-root or transversality restriction has been used. $\square$

A.4 Cube sections and correlated density conversion

Lemma A.5 (Scaled central cube section). *Under Assumption 1, suppose $N \geq 2$ and $v \in \mathbb{R}^N$ satisfies $\|v\|_2 = 1$. Then*

$$\mathcal{H}^{N-1}(v^\perp \cap [-R, R]^N) \leq \sqrt{2}(2R)^{N-1}.\tag{A.6}$$

Proof. Let $Q_N = [-1/2, 1/2]^N$ and $H = v^\perp$. Ball's cube-slicing theorem (Ball, 1986) states, in the Euclidean intrinsic-volume convention, that every central hyperplane section of this unit-volume cube satisfies

$$\mathcal{H}^{N-1}(Q_N \cap H) \leq \sqrt{2}. \quad (\text{A.7})$$

Its hypotheses hold: $N \geq 2$, and the unit vector v makes H a codimension-one Euclidean linear hyperplane through the origin. No rotation of the coefficient law is involved.

Let $D_R(y) = 2Ry$. Since $R > 0$,

$$D_R(Q_N) = [-R, R]^N, \quad D_R(H) = H,$$

and therefore

$$D_R(Q_N \cap H) = [-R, R]^N \cap H.$$

Euclidean Hausdorff measure scales under this similarity by $(2R)^{N-1}$, directly from its covering definition. Applying the inverse dilation supplies the reverse scaling equality, so

$$\begin{aligned} \mathcal{H}^{N-1}(H \cap [-R, R]^N) &= (2R)^{N-1} \mathcal{H}^{N-1}(H \cap Q_N) \\ &\leq \sqrt{2} (2R)^{N-1}. \end{aligned}$$

The dilation maps the closed cube and all of its faces exactly. The result is orientation-uniform and concerns only central, not translated, sections. $\square$

Proposition A.4 (Central sweep under a capped joint law). *Under Assumptions 1 and 4, Proposition A.3, and, when $N \geq 2$, Lemma A.5, every $\mu \in \mathcal{D}_{N,R,\kappa}$ and every interval $I \subseteq \Theta$ satisfy*

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : \langle \alpha, F(\theta) \rangle = 0] \leq A \sqrt{\frac{N}{2}} \int_I \|\gamma'_F(\theta)\|_2 d\theta. \quad (\text{A.8})$$

This includes empty, singleton, open, closed, and half-open intervals.

Proof. Fix the deterministic presentation, then an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$, and then an arbitrary interval I . Define the Borel set

$$\mathcal{E}_I := \{a \in [-R, R]^N : \exists \theta \in I, \langle a, F(\theta) \rangle = 0\}.$$

Its measurability is proved in Proposition A.3. If f_μ is the full joint density, cube support and $f_\mu \leq \kappa$ almost everywhere give the only probability conversion:

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : \langle \alpha, F(\theta) \rangle = 0] = \int_{\mathcal{E}_I} f_\mu(a) da \leq \kappa \text{Leb}^N(\mathcal{E}_I). \quad (\text{A.9})$$

This uses neither conditional densities nor coordinate independence. The nonemptiness of the law class also gives

$$1 = \int_{[-R,R]^N} f_\mu(a) da \leq \kappa (2R)^N = A,$$

although this fact is not used to relax any later coefficient.

If $N = 1$, Proposition A.3 gives $\text{Leb}^1(\mathcal{E}_I) = 0$: the section is $\{0\}$ and its integrand vanishes. Moreover the unit scalar curve has $\gamma'_F = 0$. Thus (A.8) is exactly $0 \leq 0$, including all interval conventions, and Ball's positive-dimensional theorem is not invoked.

Assume $N \geq 2$. Equations (A.3) and (A.9) give

$$\begin{aligned} & \Pr_{\alpha \sim \mu} [\exists \theta \in I : \langle \alpha, F(\theta) \rangle = 0] \\ & \leq \kappa \int_I \int_{\gamma_F(\theta)^\perp \cap [-R, R]^N} |\langle \gamma'_F(\theta), a \rangle| d\mathcal{H}^{N-1}(a) d\theta. \end{aligned} \quad (\text{A.10})$$

For $a \in [-R, R]^N$, Cauchy–Schwarz (Rudin, 1976) and cube support give

$$|\langle \gamma'_F(\theta), a \rangle| \leq \|\gamma'_F(\theta)\|_2 \|a\|_2 \leq R\sqrt{N} \|\gamma'_F(\theta)\|_2. \quad (\text{A.11})$$

Lemma A.5, applied to the unit vector $\gamma_F(\theta)$, therefore yields

$$\begin{aligned} & \int_{\gamma_F(\theta)^\perp \cap [-R, R]^N} |\langle \gamma'_F(\theta), a \rangle| d\mathcal{H}^{N-1}(a) \\ & \leq R\sqrt{N} \|\gamma'_F(\theta)\|_2 \mathcal{H}^{N-1}(\gamma_F(\theta)^\perp \cap [-R, R]^N) \\ & \leq R\sqrt{N} \sqrt{2}(2R)^{N-1} \|\gamma'_F(\theta)\|_2. \end{aligned} \quad (\text{A.12})$$

Only a central linear section is used. Substitution into (A.10) gives the coefficient

$$\begin{aligned} \kappa R\sqrt{N} \sqrt{2}(2R)^{N-1} &= \kappa(2R)^N \frac{\sqrt{2N}}{2} \\ &= \kappa(2R)^N \sqrt{\frac{N}{2}} = A\sqrt{\frac{N}{2}}, \end{aligned} \quad (\text{A.13})$$

where $R(2R)^{N-1} = (2R)^N/2$. This is equality, not a term absorption, and proves (A.8). The upper bound need not be at most one and is not clipped.

Endpoint inclusion does not alter the argument. Proposition A.3 already accounts for endpoint hyperplanes, tangent and multiple roots, stationary pieces, infinite fibers, and identically zero combinations. A stationary curve makes its incidence integral and event probability zero. The dilation in Lemma A.5 changes only deterministic section measure; it neither transforms the coefficient vector nor its density. Since the law and interval were arbitrary after the deterministic presentation was fixed, the required quantifier order follows without a union bound over laws, intervals, roots, or orientations. $\square$

A.5 Central rates and ordered suprema

Lemma A.6 (Projective speed controls interval length). *Under Assumption 1 and Proposition A.2, every interval $I \subseteq \Theta$, with any endpoint convention, satisfies*

$$\int_I \|\gamma'_F(\theta)\|_2 d\theta \leq \Gamma_{\text{proj}}(F)|I|. \quad (\text{A.14})$$

Proof. Proposition A.2 gives $\gamma_F \in C^1(\Theta; \mathbb{R}^N)$, so $\theta \mapsto \|\gamma'_F(\theta)\|_2$ is nonnegative and measurable. By the definition of the essential supremum, outside a null subset of Θ ,

$$\|\gamma'_F(\theta)\|_2 \leq \Gamma_{\text{proj}}(F).$$

Integration over I gives

$$\int_I \|\gamma'_F(\theta)\|_2 d\theta \leq \int_I \Gamma_{\text{proj}}(F) d\theta = \Gamma_{\text{proj}}(F)|I|.$$

Changing endpoint inclusion affects neither the integral nor the length, so the assertion also includes empty and singleton intervals. $\square$

Proposition A.5 (Central rate bridge and supremum closure). *Under Assumptions 1, 2, 3, and 4, Propositions A.2 and A.4, and Lemma A.6, every $\mu \in \mathcal{D}_{N,R,\kappa}$ and every positive-length interval $I \subseteq \Theta$ satisfy*

$$\begin{aligned} & \Pr_{\alpha \sim \mu} [\exists \theta \in I : \langle \alpha, F(\theta) \rangle = 0] \\ & \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) |I| \\ & \leq \frac{AN \Delta B_Q (1 + q B_P)}{\sqrt{2} h} |I|. \end{aligned} \quad (\text{A.15})$$

Moreover,

$$C_D^{\text{Pf}}(F; \Theta) \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) \leq \frac{AN \Delta B_Q (1 + q B_P)}{\sqrt{2} h}. \quad (\text{A.16})$$

The last coefficient is literally

$$\frac{(2R)^N \kappa N \Delta B_Q (1 + q B_P)}{\sqrt{2} h}, \quad (\text{A.17})$$

has degree-zero dependence on M , reduces to $AN \Delta B_Q / (\sqrt{2} h)$ when $q = 0$, and becomes $\sqrt{2}/\delta$ when $N = 2$, $A = h = \Delta = 1$, $q = B_P = 0$, and $B_Q = 1/\delta$.

Proof. Fix the deterministic presentation and full parameter tuple first. Then fix an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$ and an arbitrary interval I , initially allowing zero length. Define

$$Z_I := \{\alpha \in \mathbb{R}^N : \exists \theta \in I, \langle \alpha, F(\theta) \rangle = 0\}.$$

Proposition A.4 and Lemma A.6 give

$$\begin{aligned} \Pr_{\alpha \sim \mu}(Z_I) & \leq A \sqrt{\frac{N}{2}} \int_I \|\gamma'_F(\theta)\|_2 d\theta \\ & \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) |I|. \end{aligned} \quad (\text{A.18})$$

Proposition A.2 applies to the same deterministic feature and Euclidean norm. Multiplying its bound by the nonnegative factor $A \sqrt{N/2} |I|$ gives

$$\begin{aligned} A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) |I| & \leq A \sqrt{\frac{N}{2}} \frac{\sqrt{N} \Delta B_Q (1 + q B_P)}{h} |I| \\ & = \frac{AN \Delta B_Q (1 + q B_P)}{\sqrt{2} h} |I|, \end{aligned} \quad (\text{A.19})$$

where the complete simplification is

$$\sqrt{\frac{N}{2}} \sqrt{N} = \sqrt{\frac{N^2}{2}} = \frac{N}{\sqrt{2}} \quad (N \geq 1). \quad (\text{A.20})$$

No term is dropped. Equations (A.18)–(A.20) prove (A.15). If I is empty or a singleton, the speed integral and $|I|$ are zero, so the event has probability zero before any quotient. Endpoint variants have the same length and integral, while Proposition A.4 retains the root-event boundary handling.

Now restrict to $|I| > 0$ and divide (A.15) by this positive length:

$$\frac{\Pr_{\alpha \sim \mu}(Z_I)}{|I|} \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) \leq \frac{AN \Delta B_Q (1 + q B_P)}{\sqrt{2} h}. \quad (\text{A.21})$$

For fixed μ , first take the interval supremum. The two right sides are independent of I , so

$$\sup_{\substack{I \subseteq \Theta \text{ interval} \\ |I| > 0}} \frac{\Pr_{\alpha \sim \mu}(Z_I)}{|I|} \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) \leq \frac{AN \Delta B_Q (1 + qB_P)}{\sqrt{2}h}. \quad (\text{A.22})$$

Only then take the law supremum:

$$\begin{aligned} & \sup_{\mu \in \mathcal{D}_{N,R,\kappa}} \sup_{\substack{I \subseteq \Theta \text{ interval} \\ |I| > 0}} \frac{\Pr_{\alpha \sim \mu}(Z_I)}{|I|} \\ & \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) \leq \frac{AN \Delta B_Q (1 + qB_P)}{\sqrt{2}h}. \end{aligned} \quad (\text{A.23})$$

The left side is the definition of $C_{\mathcal{D}}^{\text{Pf}}(F; \Theta)$, proving (A.16).

Substituting $A = (2R)^N \kappa$ gives (A.17) by equality. The parameter M does not occur, so its dependence is degree zero for fixed B_P . At $q = 0$, $M = B_P = 0$ and $1 + qB_P = 1$. For $N = 2$, $A = h = \Delta = 1$, $q = B_P = 0$, and $B_Q = 1/\delta$, the coefficient is

$$\frac{1 \cdot 2 \cdot 1 \cdot (1/\delta) \cdot 1}{\sqrt{2} \cdot 1} = \frac{\sqrt{2}}{\delta}. \quad (\text{A.24})$$

If $\Gamma_{\text{proj}}(F) = 0$, (A.18) gives zero probability for every law and interval, hence zero capacity; no division by speed occurs. When $N = 1$, the anchor gives $F = (1)$, $\Delta = 0$, and all displayed chains reduce to $0 \leq 0$. The only division in the closure is by the explicitly positive interval length. $\square$

A.6 Finite affine charts and measurable exhaustion

Lemma A.7 (Finite exhausted-chart legality). *Under Assumptions 1, 2, 3, and 5, and Proposition A.1, fix an interval $I \subseteq \Theta$, an index $1 \leq j \leq N$, and $m \geq 1$. Put*

$$E_{j,m} := \{\theta \in E_j : |F_j(\theta)| \geq 1/m\}.$$

Index $\beta \in [-R, R]^{N-1}$ by $i \neq j$, using the zero-dimensional singleton convention at $N = 1$, and define

$$\Psi_j(\theta, \beta)_i = \begin{cases} \beta_i, & i \neq j, \\ T_j(\theta, \beta), & i = j. \end{cases}$$

Then T_j and Ψ_j are measurable and Lipschitz on $E_{j,m} \times [-R, R]^{N-1}$. Their restrictions to measurable subsets satisfy the regularity hypotheses of the equal-dimensional area formula. No Lipschitz constant uniform in m is asserted.

Proof. For every $\theta \in \Theta$, the unit-box coefficient estimate gives

$$|F_i(\theta)| = |Q_i(x(\theta), \eta(x(\theta)))| \leq \|\text{coeff}(Q_i)\|_1 \leq B_Q. \quad (\text{A.25})$$

Proposition A.1 and the affine coordinate change give

$$|F'_i(\theta)| \leq L_F := D_*/h. \quad (\text{A.26})$$

The proof-local quantities

$$B_H := \|F_0\|_\infty + (N-1)RB_Q, \quad L_H := \|F'_0\|_\infty + (N-1)RL_F \quad (\text{A.27})$$

are finite because $F_0 \in C^1(\Theta)$, Θ is compact, and the remaining terms are setting quantities or the bound (A.26). Define

$$H_j(\theta, \beta) := F_0(\theta) + \sum_{i \neq j} \beta_i F_i(\theta), \quad T_j = -H_j/F_j.$$

Then

$$|H_j(\theta, \beta)| \leq B_H,$$

and, for $\theta, s \in \Theta$ and $\beta, \zeta \in [-R, R]^{N-1}$,

$$|H_j(\theta, \beta) - H_j(s, \zeta)| \leq L_H |\theta - s| + B_Q \|\beta - \zeta\|_1. \quad (\text{A.28})$$

If $\theta, s \in E_{j,m}$, both denominators have magnitude at least $1/m$. The quotient identity, (A.26), and (A.28) yield

$$\begin{aligned} |T_j(\theta, \beta) - T_j(s, \zeta)| &\leq m |H_j(\theta, \beta) - H_j(s, \zeta)| \\ &\quad + m^2 |H_j(s, \zeta)| |F_j(\theta) - F_j(s)| \\ &\leq (mL_H + m^2 B_H L_F) |\theta - s| + mB_Q \|\beta - \zeta\|_1. \end{aligned} \quad (\text{A.29})$$

Thus T_j is Lipschitz even if the measurable finite-level set is disconnected; inserting the unchanged beta coordinates makes Ψ_j Lipschitz. At interior points,

$$|\partial_\theta T_j(\theta, \beta)| \leq mL_H + m^2 B_H L_F, \quad |\partial_{\beta_i} T_j(\theta, \beta)| \leq mB_Q. \quad (\text{A.30})$$

The same bounds hold one-sided at ambient endpoints, and the finite-level Jacobian is bounded on its finite product domain.

On the open nonzero-pivot locus in $\text{int } \Theta$, the quotient formulas are C^1 . Hence at almost every density point of a measurable restricted domain, the approximate derivative of the Lipschitz restriction agrees with this C^1 derivative. The ambient endpoints are null affine slices, as established in Lemma A.9 below. Measurability follows from measurability of E_j and continuity of the quotient on the nonzero-pivot locus. When $N = 1$, all beta sums and norms are empty and the argument is the same scalar quotient estimate. $\square$

Lemma A.8 (Exact affine coefficient chart and Jacobian). *Under the assumptions and fixed objects of Lemma A.7, define*

$$D_{j,m} := \{(\theta, \beta) : \theta \in E_{j,m}, \beta \in [-R, R]^{N-1}, |T_j(\theta, \beta)| \leq R\}.$$

Then $D_{j,m}$ is measurable, $\Psi_j(D_{j,m}) \subseteq [-R, R]^N$, and every point in $D_{j,m}$ satisfies

$$F_0(\theta) + \langle \Psi_j(\theta, \beta), F(\theta) \rangle = 0. \quad (\text{A.31})$$

Conversely, if $\alpha \in [-R, R]^N$ has a root at $\theta \in E_{j,m}$, then, with $\beta = \alpha_{-j}$, $(\theta, \beta) \in D_{j,m}$ and $\Psi_j(\theta, \beta) = \alpha$. At every interior chart point,

$$|\det D\Psi_j(\theta, \beta)| = |\partial_\theta T_j(\theta, \beta)|. \quad (\text{A.32})$$

At a represented root,

$$\frac{d}{d\theta} (F_0(\theta) + \langle \alpha, F(\theta) \rangle) = -F_j(\theta) \partial_\theta T_j(\theta, \beta), \quad (\text{A.33})$$

so tangent roots have zero chart Jacobian.

Proof. Measurability follows from Lemma A.7. The domain definition places every β_i and the inserted coordinate T_j in $[-R, R]$, proving cube membership. By the chart definition,

$$\begin{aligned} F_0(\theta) + \langle \Psi_j(\theta, \beta), F(\theta) \rangle &= F_0(\theta) + \sum_{i \neq j} \beta_i F_i(\theta) + T_j(\theta, \beta) F_j(\theta) \\ &= H_j(\theta, \beta) - H_j(\theta, \beta) = 0. \end{aligned}$$

Conversely, if the affine expression vanishes and $F_j(\theta) \neq 0$, solving for α_j gives $\alpha_j = T_j(\theta, \alpha_{-j})$. Cube membership of α supplies $\alpha_{-j} \in [-R, R]^{N-1}$ and $|T_j| \leq R$, proving the converse.

Permute output coordinates so coordinate j is first, and retain domain coordinates $(\theta, (\beta_i)_{i \neq j})$. The derivative matrix is

$$\begin{pmatrix} \partial_\theta T_j & (\partial_{\beta_i} T_j)_{i \neq j} \\ 0 & I_{N-1} \end{pmatrix}.$$

The coordinate permutation has determinant of magnitude one, proving (A.32). For $N = 1$, the lower-right block is the 0×0 identity with determinant one, and the formula becomes $|\Psi'_1| = |T'_1|$.

Differentiate $H_j(\theta, \beta) + T_j(\theta, \beta) F_j(\theta) = 0$ with beta fixed. At $\alpha = \Psi_j(\theta, \beta)$ this gives

$$F'_0(\theta) + \sum_{i \neq j} \alpha_i F'_i(\theta) + \alpha_j F'_j(\theta) = -F_j(\theta) \partial_\theta T_j(\theta, \beta),$$

which is (A.33). Since the pivot is nonzero, tangency is equivalent to a zero chart derivative. $\square$

Lemma A.9 (Null affine endpoint and degeneracy classes). *Under Assumptions 3, 4, and 5, let $I \subseteq \Theta$ be nonempty. For every fixed $\theta_0 \in I$,*

$$\mathcal{H}_{\theta_0} := \{\alpha \in \mathbb{R}^N : F_0(\theta_0) + \langle \alpha, F(\theta_0) \rangle = 0\}$$

is a proper affine hyperplane and has probability zero under every $\mu \in \mathcal{D}_{N,R,\kappa}$. Hence coefficients having only an included endpoint root are law-null. The set

$$\mathcal{Z}_I := \{\alpha : F_0(\theta) + \langle \alpha, F(\theta) \rangle = 0 \text{ for every } \theta \in I\}$$

is empty or a proper affine subspace and is also law-null.

Proof. The anchor gives $F_1(\theta_0) = 1$, so

$$\alpha_1 = -F_0(\theta_0) - \sum_{i=2}^N \alpha_i F_i(\theta_0)$$

is a proper affine graph. For each fixed $(\alpha_2, \dots, \alpha_N)$, its fiber in the first coordinate is a single point and is one-dimensional Lebesgue-null. Tonelli's theorem (Bogachev, 2007) gives $\text{Leb}^N(\mathcal{H}_{\theta_0}) = 0$; at $N = 1$, the empty-sum graph is itself one point. Absolute continuity gives law nullity.

There are at most two included endpoints, so their root classes lie in a finite union of such hyperplanes. Moreover $\mathcal{Z}_I = \bigcap_{\theta \in I} \mathcal{H}_\theta$ is empty or affine, and for any $\theta_0 \in I$ it lies in the proper null hyperplane $\mathcal{H}_{\theta_0}$. If I is empty, the root event is empty and no degenerate exception is needed. Only the full joint absolute continuity has been used. $\square$

Proposition A.6 (Finite-level chart area and density bound). *Under Assumptions 1, 2, 3, 4, and 5, Proposition A.1, and Lemmas A.7, A.8, and A.9, fix an interval I , $m \geq 1$, and $\mu \in \mathcal{D}_{N,R,\kappa}$. Define*

$$\mathcal{B}_{j,m} := \Psi_j(D_{j,m}), \quad \mathcal{R}_m := \bigcup_{j=1}^N \mathcal{B}_{j,m}.$$

These coefficient sets are measurable in their Lebesgue and μ completions, and

$$\begin{aligned} \Pr_{\alpha \sim \mu} [\alpha \in \mathcal{R}_m] &\leq \kappa \sum_{j=1}^N \int_{E_{j,m}} \int_{[-R,R]^{N-1}} \mathbf{1}_{\{|T_j(\theta,\beta)| \leq R\}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta \\ &\leq \kappa \sum_{j=1}^N \int_{E_{j,m}} \int_{[-R,R]^{N-1}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta. \end{aligned} \quad (\text{A.34})$$

Both finite-level right sides are finite. Injectivity and a bound on root multiplicity are unnecessary.

Proof. If I is empty, every set and integral in (A.34) is empty, so assume I is nonempty. Fix j . Lemma A.7 makes Ψ_j Lipschitz on the finite-level domain containing $D_{j,m}$. Away from the at most two endpoints of Θ , its approximate derivative agrees almost everywhere with the C^1 chart derivative. Endpoint slices map into the null hyperplanes in Lemma A.9. The equal-dimensional area formula (Federer, 1969) and Lemma A.8 therefore give

$$\begin{aligned} \text{Leb}^N(\mathcal{B}_{j,m}) &\leq \int_{D_{j,m}} |\det D\Psi_j(\theta, \beta)| d\beta d\theta \\ &= \int_{E_{j,m}} \int_{[-R,R]^{N-1}} \mathbf{1}_{\{|T_j(\theta,\beta)| \leq R\}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta. \end{aligned} \quad (\text{A.35})$$

The formula applies to a Lipschitz extension of the map from its measurable domain. If that domain is not Borel, write it as a Borel set plus a null set. The equal-dimensional Lipschitz image of the null part is null, while the Borel image is analytic and hence universally measurable (Bogachev, 2007). This proves the completed-measure assertion without strengthening the assumed measurability of E_j .

The area formula integrates the full preimage-counting function. Multiple roots can only increase the right side of (A.35). By Lemma A.8, tangent roots lie in the zero-Jacobian part, whose image is null under the same formula. Affine-identically-zero coefficients are independently law-null by Lemma A.9; neither root simplicity nor transversality is used.

Every $\mathcal{B}_{j,m}$ lies in $[-R, R]^N$. Finite subadditivity and the single full joint-density cap give

$$\mu(\mathcal{R}_m) \leq \kappa \text{Leb}^N(\mathcal{R}_m) \leq \kappa \sum_{j=1}^N \text{Leb}^N(\mathcal{B}_{j,m}). \quad (\text{A.36})$$

Combining (A.35)–(A.36) proves the first line of (A.34); removing the indicator proves the second by nonnegativity. The finite-level derivative bounds (A.30) and the finite measure of $I \times [-R, R]^{N-1}$ make both right sides finite. At $N = 1$, integration over $[-R, R]^0$ has mass one and the same calculation is one-dimensional. $\square$

Lemma A.10 (Least-level pivot exhaustion). *Under Assumptions 3 and 5, and Lemma A.8, define*

$$\mathcal{R}_I := \{\alpha \in [-R, R]^N : \exists \theta \in I, F_0(\theta) + \langle \alpha, F(\theta) \rangle = 0\}.$$

With $\mathcal{R}_m$ as in Proposition A.6, let

$$C_{j,m} := E_{j,m} \setminus E_{j,m-1}, \quad E_{j,0} := \emptyset.$$

Then

$$E_{j,m} = \bigsqcup_{\ell=1}^m C_{j,\ell},$$

the cells are measurable, and every $\theta \in E_j$ has a unique least finite activation level. If

$$\widehat{D}_{j,m} := D_{j,m} \cap (C_{j,m} \times [-R, R]^{N-1}),$$

then

$$D_{j,m} = \bigsqcup_{\ell=1}^m \widehat{D}_{j,\ell}, \quad \mathcal{R}_m \subseteq \mathcal{R}_{m+1}, \quad \mathcal{R}_I = \bigcup_{m=1}^{\infty} \mathcal{R}_m. \quad (\text{A.37})$$

The equality includes endpoint, tangent, multiple, and identically-zero roots.

Proof. Since $1/(m+1) \leq 1/m$, the measurable sets satisfy $E_{j,m} \subseteq E_{j,m+1}$. The pairwise disjoint differences telescope to $E_{j,m} = \bigsqcup_{\ell=1}^m C_{j,\ell}$. Intersecting with the measurable restriction $|T_j| \leq R$ gives the stated disjoint identity for $D_{j,m}$.

For $\theta \in E_j$, the selected pivot is nonzero. Therefore

$$\{m \geq 1 : |F_j(\theta)| \geq 1/m\}$$

is a nonempty terminal subset of the positive integers and has the unique least element

$$m_*(\theta) := \min\{m \geq 1 : |F_j(\theta)| \geq 1/m\}. \quad (\text{A.38})$$

Equivalently, $\theta \in C_{j,m_*(\theta)}$. The partition $(E_j)_{j=1}^N$ also assigns a unique chart owner, so $(j, m_*(\theta))$ is the unique least finite chart label. No lower bound uniform in θ or m has been introduced.

The set inclusions imply $D_{j,m} \subseteq D_{j,m+1}$, and the same Ψ_j is used at every level, hence $\mathcal{R}_m \subseteq \mathcal{R}_{m+1}$. Each chart image consists of root coefficients by Lemma A.8, proving $\bigcup_m \mathcal{R}_m \subseteq \mathcal{R}_I$.

Conversely, let $\alpha \in \mathcal{R}_I$ have a witnessing root θ . The partition assigns θ to a unique E_j , and its pivot is nonzero. At the least level (A.38), put $\beta = \alpha_{-j}$. The converse part of Lemma A.8 gives $(\theta, \beta) \in D_{j,m_*(\theta)}$ and $\Psi_j(\theta, \beta) = \alpha$. Thus $\alpha \in \mathcal{R}_{m_*(\theta)}$, proving (A.37).

For measurability, $\mathcal{R}_I$ is the coefficient projection of the zero set of a continuous function on the Borel set $I \times [-R, R]^N$; it is analytic and universally measurable (Bogachev, 2007). The finite chart events have the completed measurability from Proposition A.6. Endpoint and degenerate classes are not removed from (A.37); their nullity is a separate measure statement. If I is empty, both sides are empty. $\square$

Proposition A.7 (General affine chart inequality). *Under Assumptions 1, 2, 3, 4, and 5, Proposition A.1, Proposition A.6, and Lemma A.10, every $\mu \in \mathcal{D}_{N,R,\kappa}$ and every interval $I \subseteq \Theta$ satisfy*

$$\begin{aligned} & \Pr_{\alpha \sim \mu} [\exists \theta \in I : F_0(\theta) + \langle \alpha, F(\theta) \rangle = 0] \\ & \leq \kappa \sum_{j=1}^N \int_{E_j} \int_{[-R,R]^{N-1}} \mathbf{1}_{\{|T_j(\theta,\beta)| \leq R\}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta \\ & \leq \kappa \sum_{j=1}^N \int_{E_j} \int_{[-R,R]^{N-1}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta. \end{aligned} \quad (\text{A.39})$$

Both inequalities hold in $[0, +\infty]$.

Proof. The set equality in Lemma A.10 and continuity from below (Bogachev, 2007) give

$$\mu(\mathcal{R}_I) = \lim_{m \rightarrow \infty} \mu(\mathcal{R}_m). \quad (\text{A.40})$$

Because every admissible law is supported on the coefficient cube, the left side is precisely the affine root-event probability.

For each j , the nonnegative measurable functions

$$\mathbf{1}_{E_{j,m}}(\theta) \mathbf{1}_{\{|T_j(\theta,\beta)| \leq R\}} |\partial_\theta T_j(\theta, \beta)|$$

increase pointwise on $E_j \times [-R, R]^{N-1}$ to the same expression with E_j in place of $E_{j,m}$. Apply Proposition A.6, then monotone convergence (Bogachev, 2007) and finiteness of the chart index set:

$$\begin{aligned} \mu(\mathcal{R}_I) &\leq \kappa \lim_{m \rightarrow \infty} \sum_{j=1}^N \int_{E_{j,m}} \int_{[-R,R]^{N-1}} \mathbf{1}_{\{|T_j| \leq R\}} |\partial_\theta T_j| d\beta d\theta \\ &= \kappa \sum_{j=1}^N \int_{E_j} \int_{[-R,R]^{N-1}} \mathbf{1}_{\{|T_j| \leq R\}} |\partial_\theta T_j| d\beta d\theta. \end{aligned} \tag{A.41}$$

All limits and integrals are in $[0, +\infty]$, so no finiteness of the limiting chart integral is required. If it diverges, the right side is $+\infty$. Dropping the indicator by nonnegativity proves the second inequality in (A.39). The density cap was used only on the full coefficient set in Proposition A.6; no coordinate conditioning was introduced. The preceding results include empty and singleton intervals, endpoints, $N = 1$, near-zero pivots, tangent and multiple roots, infinite fibers, and identically-zero affine combinations. $\square$

A.7 Exact monic presentation and chart velocities

Proposition A.8 (Exact normalized monic presentation). *Under Assumptions 1, 2, and 3, let $d \geq 1$ and let $J \subset \mathbb{R}$ be a bounded interval. For the specialization with $N = d$, there is a nondegenerate interval $\Theta = [c - h, c + h] \supseteq J$ for which*

$$F_0(\theta) = \theta^d, \quad F_{k+1}(\theta) = \theta^k \quad (0 \leq k \leq d-1), \tag{A.42}$$

has normalized-coordinate representatives

$$Q_0(x) = (c + hx)^d, \quad Q_{k+1}(x) = (c + hx)^k \quad (0 \leq k \leq d-1). \tag{A.43}$$

For every $\theta \in \Theta$ and $\alpha = (\alpha_0, \dots, \alpha_{d-1}) \in \mathbb{R}^d$,

$$F_0(\theta) + \langle \alpha, F(\theta) \rangle = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k = p_\alpha(\theta). \tag{A.44}$$

The specialization tuple is

$$q = 0, \quad M = 0, \quad B_P = 0, \quad N = d, \quad A = (2R)^d \kappa, \quad \Delta_{\text{aug}} = d. \tag{A.45}$$

The leading coefficient 1 is deterministic and is not a coordinate of the random lower-coefficient vector.

Proof. Since J is bounded, choose finite $L < U$ with $J \subseteq [L, U]$ and put

$$c := \frac{L+U}{2}, \quad h := \frac{U-L}{2} > 0.$$

Then $\Theta = [L, U]$ contains J . Empty and singleton intervals are included by choosing a strict margin around them.

Set $q = 0$. The chain is empty, so the conventions give $M = B_P = 0$ and $G_i = Q_i$. For $x(\theta) = (\theta - c)/h$,

$$c + hx(\theta) = c + h \frac{\theta - c}{h} = \theta.$$

This proves (A.42)–(A.43). In particular $Q_1(x) = (c + hx)^0 = 1$, so the anchor holds and the chain-range clause is vacuous. All outputs are polynomials, hence $F_0 \in C^1(\Theta)$, and their monomial coefficient budgets are finite static presentation data.

There are exactly d random-feature outputs, indexed by $k = 0, \dots, d-1$, so $N = d$ and

$$A = (2R)^N \kappa = (2R)^d \kappa.$$

Because $h > 0$, the coefficient of x^d in $(c + hx)^d$ is $h^d \neq 0$, so $\deg Q_0 = d$, while $\deg Q_{k+1} = k \leq d-1$. The same affine coordinate change has nonzero slope and preserves these degrees. Hence

$$\Delta_{\text{aug}} := \max\{\deg F_0, \deg F_1, \dots, \deg F_d\} = d.$$

Substitution gives (A.44). The leading coefficient occurs only in the deterministic offset F_0 , leaving exactly the d lower coefficients in α . $\square$

Lemma A.11 (Prescribed monic pivot partition). *Under Assumption 1 and Proposition A.8, if $d \geq 2$, then*

$$E_1 = J \cap \{|\theta| \leq 1\}, \quad E_d = J \cap \{|\theta| > 1\}, \quad E_j = \emptyset \quad (j \notin \{1, d\}) \quad (\text{A.46})$$

is a measurable partition of J , with $F_1 \neq 0$ on E_1 and $F_d \neq 0$ on E_d . If $d = 1$, the choice $E_1 = J$ is measurable and has $F_1 \neq 0$. The conclusions include $\theta = 0$, $|\theta| = 1$, and empty cells.

Proof. Intervals and the two sets $\{|\theta| \leq 1\}$ and $\{|\theta| > 1\}$ are Borel. The latter sets are disjoint and cover $\mathbb{R}$, proving the partition and its measurability. Empty-cell pivot conditions are vacuous.

Proposition A.8 gives $F_1(\theta) = \theta^0 = 1$. Thus 0 and the transition points ± 1 , when present, lie in E_1 and have pivot one. For $d \geq 2$,

$$F_d(\theta) = \theta^{d-1},$$

which is nonzero, indeed has magnitude greater than one, on E_d . At $d = 1$, the sole cell $E_1 = J$ has the same constant pivot. The construction does not depend on the location of J ; an unused regime simply gives an empty cell. $\square$

Lemma A.12 (Constant-pivot chart). *Under Assumption 1, Proposition A.8, and Lemma A.11, if $d \geq 2$, $\theta \in E_1$, and*

$$\beta = (\beta_1, \dots, \beta_{d-1}) \in [-R, R]^{d-1},$$

then

$$T_1(\theta, \beta) = -\theta^d - \sum_{k=1}^{d-1} \beta_k \theta^k, \quad (\text{A.47})$$

and

$$|\partial_\theta T_1(\theta, \beta)| \leq d + R \sum_{k=1}^{d-1} k = d + \frac{Rd(d-1)}{2}. \quad (\text{A.48})$$

If $d = 1$, then $T_1(\theta) = -\theta$ and $|T_1'(\theta)| = 1$.

Proof. For $d \geq 2$, index nonpivot coordinates by polynomial exponent. On E_1 , the root equation with α_0 as pivot is

$$0 = \theta^d + \alpha_0 + \sum_{k=1}^{d-1} \beta_k \theta^k.$$

Solving gives (A.47), and direct differentiation gives

$$\partial_\theta T_1 = -d\theta^{d-1} - \sum_{k=1}^{d-1} k\beta_k \theta^{k-1}.$$

Since $|\theta| \leq 1$ on E_1 and $|\beta_k| \leq R$,

$$\begin{aligned} |\partial_\theta T_1| &\leq d|\theta|^{d-1} + \sum_{k=1}^{d-1} k|\beta_k||\theta|^{k-1} \\ &\leq d + R \sum_{k=1}^{d-1} k = d + \frac{Rd(d-1)}{2}. \end{aligned}$$

At $\theta = 0$, the $k = 1$ term has the ordinary factor $\theta^0 = 1$ and all positive powers vanish; this chart contains no negative power. At $d = 1$, the equation is $\theta + \alpha_0 = 0$, so the beta tuple and sum are empty, $T_1 = -\theta$, and its speed is one, equal to $1 + R \cdot 1 \cdot 0/2$. $\square$

Lemma A.13 (Highest-lower-degree monic chart). *Under Assumption 1, Proposition A.8, and Lemma A.11, if $d \geq 2$, $\theta \in E_d$, and $\beta = (\beta_0, \dots, \beta_{d-2}) \in [-R, R]^{d-1}$, then*

$$T_d(\theta, \beta) = -\theta - \sum_{k=0}^{d-2} \beta_k \theta^{k-d+1}, \quad (\text{A.49})$$

and

$$|\partial_\theta T_d(\theta, \beta)| \leq 1 + R \sum_{k=0}^{d-2} (d-1-k) = 1 + \frac{Rd(d-1)}{2} \leq d + \frac{Rd(d-1)}{2}. \quad (\text{A.50})$$

All negative powers are confined to $E_d \subset \{|\theta| > 1\}$.

Proof. Index nonpivot coordinates by exponent. The active pivot is $F_d(\theta) = \theta^{d-1} \neq 0$, so the root equation

$$0 = \theta^d + \sum_{k=0}^{d-2} \beta_k \theta^k + \alpha_{d-1} \theta^{d-1}$$

can be divided by θ^{d-1} , giving (A.49). Differentiating on the strict high cell gives

$$\partial_\theta T_d = -1 + \sum_{k=0}^{d-2} (d-1-k) \beta_k \theta^{k-d}.$$

Here $k-d < 0$ and $|\theta| > 1$, hence $|\theta^{k-d}| \leq 1$. Thus

$$\begin{aligned} |\partial_\theta T_d| &\leq 1 + \sum_{k=0}^{d-2} (d-1-k) |\beta_k| |\theta|^{k-d} \\ &\leq 1 + R \sum_{k=0}^{d-2} (d-1-k) \\ &= 1 + \frac{Rd(d-1)}{2}. \end{aligned}$$

Since $d \geq 2$, $1 \leq d$, proving (A.50).

At $d = 2$, the two formulas reduce visibly to

$$T_1 = -\theta^2 - \beta_1\theta, \quad |T'_1| \leq 2 + R \quad (|\theta| \leq 1),$$

and

$$T_2 = -\theta - \beta_0\theta^{-1}, \quad T'_2 = -1 + \beta_0\theta^{-2}, \quad |T'_2| \leq 1 + R \leq 2 + R \quad (|\theta| > 1).$$

The negative powers are never evaluated at 0 or $|\theta| = 1$; those points belong to the constant-pivot cell. $\square$

A.8 Lossless affine-monic specialization

Proposition A.9 (Lossless monic transfer through the affine bound). *Under Assumptions 1 and 4, Proposition A.7, Proposition A.8, and Lemma A.11, let $d \geq 1$, $\mu \in \mathcal{D}_{d,R,\kappa}$, and let $J \subset \mathbb{R}$ be a bounded interval. For the monic presentation and pivot cells in Proposition A.8 and Lemma A.11,*

$$\begin{aligned} & \Pr_{\alpha \sim \mu} [\exists \theta \in J : p_\alpha(\theta) = 0] \\ & \leq \kappa \sum_{j=1}^d \int_{E_j} \int_{[-R,R]^{d-1}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta. \end{aligned} \tag{A.51}$$

The random vector has exactly d coordinates, all lower coefficients, and the leading monic coefficient is deterministic.

Proof. Fix d , then an arbitrary $\mu \in \mathcal{D}_{d,R,\kappa}$, and then an arbitrary bounded J . Proposition A.8 supplies a nondegenerate deterministic $\Theta \supseteq J$, sets $N = d$, and proves the pointwise identity

$$F_0(\theta) + \langle \alpha, F(\theta) \rangle = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k = p_\alpha(\theta). \tag{A.52}$$

Thus the affine event equals, rather than contains, the polynomial-root event. The same proposition supplies deterministic C^1 data and keeps $\alpha = (\alpha_0, \dots, \alpha_{d-1}) \in \mathbb{R}^d$ without a leading coordinate.

Lemma A.11 supplies the measurable partition of J and proves that every selected pivot is nonzero. All interval, dimension, object, regularity, and pivot hypotheses of Proposition A.7 therefore hold with $I = J$ and $N = d$. Assumption 4 supplies its one full d -dimensional density cap, with arbitrary correlation. Applying that proposition gives (A.51). No chart union bound, coordinate conditioning, law factorization, or root count is added; cube faces and endpoint conventions remain in the affine interface. $\square$

Lemma A.14 (Coefficient-one two-cell ledger). *Under Assumption 1 and Lemmas A.11, A.12, and A.13, if $d \geq 2$ and J is bounded, then*

$$\sum_{j=1}^d \int_{E_j} \int_{[-R,R]^{d-1}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta \leq (2R)^{d-1} \left(d + \frac{Rd(d-1)}{2} \right) |J|. \tag{A.53}$$

The sharper high-chart speed is retained until its coefficient-one domination; no chart-count factor occurs.

Proof. Define within this proof

$$V_d := d + \frac{Rd(d-1)}{2}, \quad W_d := 1 + \frac{Rd(d-1)}{2}.$$

The closed beta cube has the exact product volume

$$\text{Leb}^{d-1}([-R, R]^{d-1}) = \prod_{\ell=1}^{d-1} (2R) = (2R)^{d-1}. \quad (\text{A.54})$$

Cube faces are included and do not alter either this identity or the pointwise velocity estimates.

All cells other than E_1 and E_d are empty. The two chart lemmas give

$$\int_{E_1} \int_{[-R, R]^{d-1}} |\partial_\theta T_1| d\beta d\theta \leq (2R)^{d-1} V_d |E_1|, \quad (\text{A.55})$$

$$\int_{E_d} \int_{[-R, R]^{d-1}} |\partial_\theta T_d| d\beta d\theta \leq (2R)^{d-1} W_d |E_d|. \quad (\text{A.56})$$

Neither cell length has been replaced by $|J|$. Since $d \geq 2$,

$$V_d - W_d = d - 1 \geq 0, \quad W_d \leq V_d. \quad (\text{A.57})$$

Therefore

$$\begin{aligned} \sum_{j=1}^d \int_{E_j} \int_{[-R, R]^{d-1}} |\partial_\theta T_j| d\beta d\theta \\ \leq (2R)^{d-1} (V_d |E_1| + W_d |E_d|) \\ \leq (2R)^{d-1} V_d (|E_1| + |E_d|). \end{aligned} \quad (\text{A.58})$$

The non-strict low condition assigns both transition points to E_1 and the strict complementary condition assigns every other point to E_d . Hence, for every endpoint convention,

$$|E_1| + |E_d| = |J|. \quad (\text{A.59})$$

Substituting (A.59) in (A.58) proves (A.53). If either cell is empty, its integral and length vanish. The calculation is independent of the location of J . $\square$

Proposition A.10 (Zero-dimensional linear branch). *Under Assumptions 1 and 4, Proposition A.9, Proposition A.8, and Lemmas A.11 and A.12, if $d = 1$, $\mu \in \mathcal{D}_{1,R,\kappa}$, and J is bounded, then*

$$\Pr_{\alpha_0 \sim \mu} [\exists \theta \in J : \theta + \alpha_0 = 0] \leq \kappa |J| = \kappa (2R)^0 \left(1 + \frac{R \cdot 1 \cdot 0}{2}\right) |J|. \quad (\text{A.60})$$

Proof. At $d = 1$, Proposition A.8 gives $N = 1$, $\alpha = (\alpha_0)$, and $p_\alpha(\theta) = \theta + \alpha_0$, with a deterministic coefficient of θ . The prescribed partition is $E_1 = J$, and there are no nonpivot coordinates:

$$[-R, R]^0 = \{()\}, \quad \text{Leb}^0([-R, R]^0) = 1 = (2R)^0.$$

Lemma A.12 gives $T_1(\theta) = -\theta$ and $|T'_1(\theta)| = 1$. Therefore

$$\int_{E_1} \int_{[-R, R]^0} |T'_1(\theta)| d\text{Leb}^0(\beta) d\theta = \int_J 1 d\theta = |J|. \quad (\text{A.61})$$

Proposition A.9 gives the first inequality in (A.60), and $1 + R \cdot 1 \cdot 0/2 = 1$ gives the equality. There is no high chart, second cell, or positive-dimensional cube factor. Empty and singleton intervals are included by the affine interface. $\square$

Proposition A.11 (Exact affine-monic probability bound). *Under Assumptions 1 and 4, Propositions A.9 and A.10, and Lemma A.14, every $d \geq 1$, every $\mu \in \mathcal{D}_{d,R,\kappa}$, and every bounded interval $J \subset \mathbb{R}$ satisfy*

$$\Pr_{\alpha \sim \mu} [\exists \theta \in J : p_\alpha(\theta) = 0] \leq \kappa(2R)^{d-1} \left(d + \frac{Rd(d-1)}{2} \right) |J|. \quad (\text{A.62})$$

The leading coefficient remains deterministic, and arbitrary correlation among the lower coefficients is allowed.

Proof. Fix d , then an arbitrary admissible law, and then a bounded interval with its literal endpoint convention. Proposition A.9 applies the general affine inequality to the identical monic event, the exact d -dimensional lower-coefficient vector, and the prescribed cells.

If $d \geq 2$, Lemma A.14 bounds its complete chart sum by the right side of (A.53). Its calculation first uses the sharper high-chart speed, then (A.57), and only then the disjoint length ledger (A.59), so it introduces no factor for the number of charts. Multiplication by the literal affine coefficient κ proves (A.62).

If $d = 1$, Proposition A.10 gives the same formula from zero-dimensional volume one and speed one. These cases exhaust all $d \geq 1$. The result uses ordinary probability, the original joint density, and the deterministic leading coefficient; it contains no independent root theorem, probability-mode conversion, auxiliary threshold, conservative remainder, or location-dependent constant. $\square$

A.9 Counter-example presentation and scale certificate

Proposition A.12 (Counter-example presentation and projective speed). *Under Assumptions 1, 2, 3, and 4, suppose*

$$0 < \delta \leq 1, \quad \Theta = [-1, 1], \quad Q_1(x) = 1, \quad Q_2(x) = x/\delta,$$

with $c = 0$, $h = 1$, $q = 0$, and the uniform law on $[-1, 1]^2$. Then

$$F(\theta) = \left(1, \frac{\theta}{\delta} \right), \quad \langle \alpha, F(\theta) \rangle = \alpha_1 F_1(\theta) + \alpha_2 F_2(\theta),$$

and

$$q = 0, \quad M = 0, \quad B_P = 0, \quad \Delta = 1, \quad N = 2, \quad R = 1, \quad \kappa = \frac{1}{4}, \quad A = 1, \quad B_Q = \frac{1}{\delta}, \quad h = 1. \quad (\text{A.63})$$

Moreover,

$$\gamma_F(\theta) = \frac{(\delta, \theta)}{\sqrt{\delta^2 + \theta^2}}, \quad \gamma'_F(\theta) = \frac{(-\delta\theta, \delta^2)}{(\delta^2 + \theta^2)^{3/2}}, \quad (\text{A.64})$$

and

$$\|\gamma'_F(\theta)\|_2 = \frac{\delta}{\delta^2 + \theta^2}, \quad \Gamma_{\text{proj}}(F) = \frac{1}{\delta}. \quad (\text{A.65})$$

Proof. Here $x(\theta) = \theta$. Since $q = 0$, the conventions give $M = B_P = 0$. The two coefficient norms are

$$\|\text{coeff}(Q_1)\|_1 = 1, \quad \|\text{coeff}(Q_2)\|_1 = \frac{1}{\delta},$$

and $0 < \delta \leq 1$ implies $B_Q = 1/\delta$. The output degrees are zero and one, so $\Delta = 1$, and $N = 2$. The literal first feature leaves the original coefficient ordering unchanged.

The density

$$f_{\square}(a_1, a_2) = \frac{1}{4} \mathbf{1}_{[-1, 1]^2}(a_1, a_2)$$

is supported on the square, has essential supremum $1/4$, and integrates to one. Thus $\mu_\square \in \mathcal{D}_{2,1,1/4}$ and

$$A = (2R)^N \kappa = 2^2 \cdot \frac{1}{4} = 1,$$

proving (A.63).

Multiplying numerator and denominator of $F/\|F\|_2$ by $\delta > 0$ gives the first identity in (A.64), and direct differentiation gives the second. Consequently,

$$\begin{aligned} \|\gamma'_F(\theta)\|_2 &= \frac{\sqrt{\delta^2\theta^2 + \delta^4}}{(\delta^2 + \theta^2)^{3/2}} \\ &= \frac{\delta\sqrt{\delta^2 + \theta^2}}{(\delta^2 + \theta^2)^{3/2}} = \frac{\delta}{\delta^2 + \theta^2}. \end{aligned}$$

This continuous speed is maximized at $\theta = 0$, where it is $1/\delta$; its maximum and essential supremum coincide. At $\theta = \pm 1$ its value is $\delta/(\delta^2 + 1)$, so no endpoint or finiteness condition is hidden. $\square$

Lemma A.15 (Exact closed coefficient wedges). *Under the assumptions of Proposition A.12, let $0 < \epsilon \leq \delta \leq 1$ and set*

$$t := \frac{\epsilon}{\delta} \in (0, 1].$$

Define

$$W_+(t) := \{(a_1, a_2) \in [-1, 1]^2 : 0 \leq a_2 \leq 1, -ta_2 \leq a_1 \leq 0\}, \quad (\text{A.66})$$

$$W_-(t) := \{(a_1, a_2) \in [-1, 1]^2 : -1 \leq a_2 \leq 0, 0 \leq a_1 \leq -ta_2\}. \quad (\text{A.67})$$

Then

$$\{a \in [-1, 1]^2 : \exists \theta \in [0, \epsilon], a_1 + a_2\theta/\delta = 0\} = W_+(t) \cup W_-(t), \quad (\text{A.68})$$

$W_+(t) \cap W_-(t) = \{(0, 0)\}$, and

$$\Pr_{\alpha \sim \mu_\square} [\exists \theta \in [0, \epsilon] : \alpha_1 + \alpha_2\theta/\delta = 0] = \frac{\epsilon}{4\delta}. \quad (\text{A.69})$$

Proof. For $\theta \in [0, \epsilon]$, put $s = \theta/\delta$. Then $s \in [0, t]$. If $a_2 \neq 0$, the unique candidate is

$$s_* = -\frac{a_1}{a_2},$$

and it lies in the interval precisely when

$$0 \leq -\frac{a_1}{a_2} \leq t. \quad (\text{A.70})$$

For $a_2 > 0$, this becomes $-ta_2 \leq a_1 \leq 0$; for $a_2 < 0$, it becomes $0 \leq a_1 \leq -ta_2$. These are the two wedges.

The coefficient axes make (A.68) exact, not only almost everywhere. If $a_1 = 0$ and $a_2 \neq 0$, the root is the included left endpoint $\theta = 0$, and the appropriate closed wedge contains it. If $a_2 = 0$ and $a_1 \neq 0$, no root exists and both wedges exclude it. At the origin the combination vanishes for every θ , and both wedges contain it. For nonzero opposite-sign coefficients, $|a_1| = t|a_2|$ is precisely the right endpoint $\theta = \epsilon$; strict inequality gives an interior root. Same-sign nonzero coefficients have no root for $s \geq 0$.

The complete triangle boundaries are

$$\begin{aligned} \partial W_+(t) &= \{(0, u) : 0 \leq u \leq 1\} \cup \{(-tu, u) : 0 \leq u \leq 1\} \\ &\cup \{(v, 1) : -t \leq v \leq 0\}, \end{aligned} \quad (\text{A.71})$$

$$\begin{aligned}\partial W_-(t) = & \{(0, u) : -1 \leq u \leq 0\} \cup \{(-tu, u) : -1 \leq u \leq 0\} \\ & \cup \{(v, -1) : 0 \leq v \leq t\}.\end{aligned}\tag{A.72}$$

They include the coefficient-axis edges, roots at ϵ , square-support edges, and all vertices. The triangles can overlap only at $a_2 = 0$, where their inequalities force $a_1 = 0$. Each boundary is a finite union of line segments and the overlap is one point, so all are planar-null while remaining part of the exact event.

Direct integration gives

$$\text{Leb}^2(W_+(t)) = \int_0^1 tu \, du = \frac{t}{2}, \quad \text{Leb}^2(W_-(t)) = \int_{-1}^0 t|u| \, du = \frac{t}{2}.$$

The null overlap gives

$$\text{Leb}^2(W_+(t) \cup W_-(t)) = t.\tag{A.73}$$

Multiplication by the constant density $1/4$ proves (A.69). When $\epsilon = \delta$, $t = 1$, each wedge has area $1/2$, the union has area one, and the probability is $1/4$. When $\delta = 1$, the probability is $\epsilon/4$ for every $0 < \epsilon \leq 1$. Thus all parameter boundaries and both interval endpoints are included without a limit. $\square$

Proposition A.13 (Counter-example lower certificate and upper scales). *Under Assumptions 1, 2, 3, and 4, Proposition A.5, Proposition A.12, and Lemma A.15, the Counter-example specialization with $\mathcal{D} = \mathcal{D}_{2,1,1/4}$ satisfies*

$$\frac{1}{4\delta} \leq C_{\mathcal{D}}^{\text{Pf}}(F; [-1, 1]) \leq \frac{1}{\delta} \leq \frac{\sqrt{2}}{\delta}.\tag{A.74}$$

The middle value is the central projective coefficient and the final value is the raw-presentation coefficient. No equality or optimality conclusion for the capacity is asserted.

Proof. Fix any $0 < \epsilon \leq \delta$. Proposition A.12 gives $\mu_{\square} \in \mathcal{D}_{2,1,1/4}$, and $[0, \epsilon] \subseteq [-1, 1]$ has length ϵ . By Lemma A.15,

$$\frac{\Pr_{\alpha \sim \mu_{\square}}[\exists \theta \in [0, \epsilon] : \langle \alpha, F(\theta) \rangle = 0]}{|[0, \epsilon]|} = \frac{\epsilon/(4\delta)}{\epsilon} = \frac{1}{4\delta}.\tag{A.75}$$

The capacity supremum contains this law–interval pair, so

$$C_{\mathcal{D}}^{\text{Pf}}(F; [-1, 1]) \geq \frac{1}{4\delta}.\tag{A.76}$$

For the upper comparison, Proposition A.5 and the tuple and speed in Proposition A.12 give

$$A\sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) = 1 \cdot \sqrt{\frac{2}{2}} \cdot \frac{1}{\delta} = \frac{1}{\delta},\tag{A.77}$$

$$\frac{AN\Delta B_Q(1 + qB_P)}{\sqrt{2}h} = \frac{1 \cdot 2 \cdot 1 \cdot (1/\delta) \cdot 1}{\sqrt{2} \cdot 1} = \frac{\sqrt{2}}{\delta}.\tag{A.78}$$

Combining (A.76)–(A.78) proves (A.74). At $\delta = 1$, the three scales are $1/4$, 1 , and $\sqrt{2}$; at $\epsilon = \delta$, the witness probability is $1/4$ and the quotient remains $1/(4\delta)$. Neither boundary changes the law, coefficient convention, deterministic feature, ordinary-probability mode, or Euclidean norm. $\square$

A.10 Joint central, affine, monic, and scale conclusions

Proposition A.14 (Joint anchored sweep conclusions). *Under Assumptions 1, 2, and 3, with Assumption 4 in every probabilistic clause and Assumption 5 in the general affine clause, the following conclusions hold simultaneously.*

First, with $D_* = \Delta B_Q(1 + qB_P)$,

$$|G'_i(x)| \leq D_*, \quad \|G'(x)\|_2 \leq \sqrt{N} D_*,$$

$G_1 = F_1 = 1$, and

$$\gamma'_G = \frac{(I_N - \gamma_G \gamma_G^\top) G'}{\|G\|_2}, \quad \gamma'_F(\theta) = \frac{1}{h} \gamma'_G \left(\frac{\theta - c}{h} \right), \quad \Gamma_{\text{proj}}(F) \leq \frac{\sqrt{N} D_*}{h}. \quad (\text{A.79})$$

This includes one-sided endpoints, $q = 0$, $N = 1$, $\Delta = 0$, and stationary normalized curves, with degree-zero dependence on M for fixed B_P .

Second, after fixing the deterministic presentation, every $\mu \in \mathcal{D}_{N,R,\kappa}$ and then every positive-length $I \subseteq \Theta$ satisfy

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : \langle \alpha, F(\theta) \rangle = 0] \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) |I| \leq \frac{AN \Delta B_Q (1 + qB_P)}{\sqrt{2}h} |I|, \quad (\text{A.80})$$

and

$$C_D^{\text{Pf}}(F; \Theta) \leq A \sqrt{\frac{N}{2}} \Gamma_{\text{proj}}(F) \leq \frac{AN \Delta B_Q (1 + qB_P)}{\sqrt{2}h}. \quad (\text{A.81})$$

The capacity retains the interval-then-law supremum order. Before division, empty and singleton intervals have zero right side. All endpoint, multiplicity, stationary, and identically-zero cases are included.

Third, for every affine invocation and every $\mu \in \mathcal{D}_{N,R,\kappa}$,

$$\begin{aligned} & \Pr_{\alpha \sim \mu} [\exists \theta \in I : F_0(\theta) + \langle \alpha, F(\theta) \rangle = 0] \\ & \leq \kappa \sum_{j=1}^N \int_{E_j} \int_{[-R,R]^{N-1}} |\partial_\theta T_j(\theta, \beta)| d\beta d\theta, \end{aligned} \quad (\text{A.82})$$

in $[0, +\infty]$. The original coefficient dimension, arbitrary correlation, near-zero pivots, all root multiplicities, $N = 1$, and divergent integrals remain admissible.

Fourth, for every $d \geq 1$ and bounded $J \subset \mathbb{R}$, there is a nondegenerate $\Theta = [c - h, c + h] \supseteq J$ with

$$\begin{aligned} & Q_0(x) = (c + hx)^d, \quad Q_{k+1}(x) = (c + hx)^k, \quad F_0(\theta) = \theta^d, \quad F_{k+1}(\theta) = \theta^k, \\ & p_\alpha(\theta) = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k, \quad (q, M, B_P, N, A, \Delta_{\text{aug}}) = (0, 0, 0, d, (2R)^d \kappa, d). \end{aligned} \quad (\text{A.83})$$

The monic coefficient is deterministic. For $d \geq 2$, the prescribed cells are those in (A.46), and

$$T_1 = -\theta^d - \sum_{k=1}^{d-1} \beta_k \theta^k, \quad |\partial_\theta T_1| \leq d + \frac{Rd(d-1)}{2} \quad (|\theta| \leq 1), \quad (\text{A.84})$$

$$T_d = -\theta - \sum_{k=0}^{d-2} \beta_k \theta^{k-d+1}, \quad |\partial_\theta T_d| \leq 1 + \frac{Rd(d-1)}{2} \leq d + \frac{Rd(d-1)}{2} \quad (|\theta| > 1). \quad (\text{A.85})$$

At $d = 1$, $E_1 = J$, $T_1 = -\theta$, and $|T'_1| = 1$. Every possibly correlated $\mu \in \mathcal{D}_{d,R,\kappa}$ satisfies

$$\Pr_{\alpha \sim \mu} [\exists \theta \in J : p_\alpha(\theta) = 0] \leq \kappa(2R)^{d-1} \left(d + \frac{Rd(d-1)}{2} \right) |J|. \quad (\text{A.86})$$

Fifth, for $0 < \delta \leq 1$, the uniform-square specialization has

$$(h, q, M, \Delta, N, R, \kappa, A, B_P, B_Q) = \left(1, 0, 0, 1, 2, 1, \frac{1}{4}, 1, 0, \frac{1}{\delta} \right),$$

$$\gamma_F(\theta) = \frac{(\delta, \theta)}{\sqrt{\delta^2 + \theta^2}}, \quad \|\gamma'_F(\theta)\|_2 = \frac{\delta}{\delta^2 + \theta^2}, \quad \Gamma_{\text{proj}}(F) = \frac{1}{\delta}. \quad (\text{A.87})$$

For every $0 < \epsilon \leq \delta$,

$$\Pr[\exists \theta \in [0, \epsilon] : \alpha_1 + \alpha_2 \theta / \delta = 0] = \frac{\epsilon}{4\delta}, \quad (\text{A.88})$$

and therefore

$$\frac{1}{4\delta} \leq C_{\mathcal{D}_{2,1,1/4}}^{\text{Pf}}(F; [-1, 1]) \leq \frac{1}{\delta} \leq \frac{\sqrt{2}}{\delta}. \quad (\text{A.89})$$

The last chain does not claim equality or optimality for the capacity.

All displayed constants are literal and have no hidden dependence. The probability mode is ordinary probability under a fixed full joint law; the central horizon mode is per interval followed only where displayed by the interval supremum and then the law supremum. Central vector and operator norms are Euclidean, chart velocities are scalar, and coefficient and interval measures use the conventions in Section 2. No confidence parameter, independence assumption, asymptotic horizon, or normalization beyond the declared anchored setting is present.

Proof. Proposition A.1 gives the coordinatewise part of (A.79), including the endpoint and degree-zero branches. Proposition A.2 gives anchor-derived nonvanishing, the normalized-projector identity, the exact coordinate change, and the remaining projective-speed assertions in (A.79).

Proposition A.5 gives (A.80)–(A.81) with the same deterministic feature, ordinary-probability mode, literal constants, and ordered suprema. Proposition A.7 gives (A.82) in the original coefficient dimension and extended-real mode.

Proposition A.8 gives the objects and tuple in (A.83). Lemma A.11 gives the prescribed cells, and Lemmas A.12 and A.13 give (A.84)–(A.85), including the transition behavior and the zero-dimensional convention. Proposition A.11 supplies only the probability conclusion (A.86), on the same monic object and lower-coefficient law.

Proposition A.12 gives the tuple and speed in (A.87), Lemma A.15 gives (A.88) with all closed boundaries, and Proposition A.13 gives (A.89). Each clause is therefore a conclusion of the named result in the identical object, dimension, law, interval, norm, and probability mode. Their logical conjunction proves the proposition without an additional estimate, assumption, probability conversion, or mode change. $\square$

A.11 Proof of the Main Theorem

Proof of Theorem 3.1. The primitive derivative and projective clauses follow from Propositions A.1 and A.2. The central probability and capacity clauses, including the literal algebraic bridge and the interval-then-law supremum order, are Proposition A.5. The general affine clause is Proposition A.7.

For the detailed monic interface, use Proposition A.8, Lemma A.11, and Lemmas A.12 and A.13; the exact monic probability rate is Proposition A.11, whose two dimensional branches are implemented by Lemma A.14 and Proposition A.10.

For Counter-example 1, Proposition [A.12](#) supplies the tuple and Euclidean speed, Lemma [A.15](#) supplies the closed-event probability, and Proposition [A.13](#) supplies the three distinct scale comparisons. Proposition [A.14](#) verifies that these named conclusions have the same assumption bases, objects, dimensions, law and interval scopes, norm conventions, and ordinary-probability mode as the clauses in Theorem [3.1](#). Their direct conjunction is precisely Theorem [3.1](#). $\square$